

Frequency Dividers and Ring Oscillators

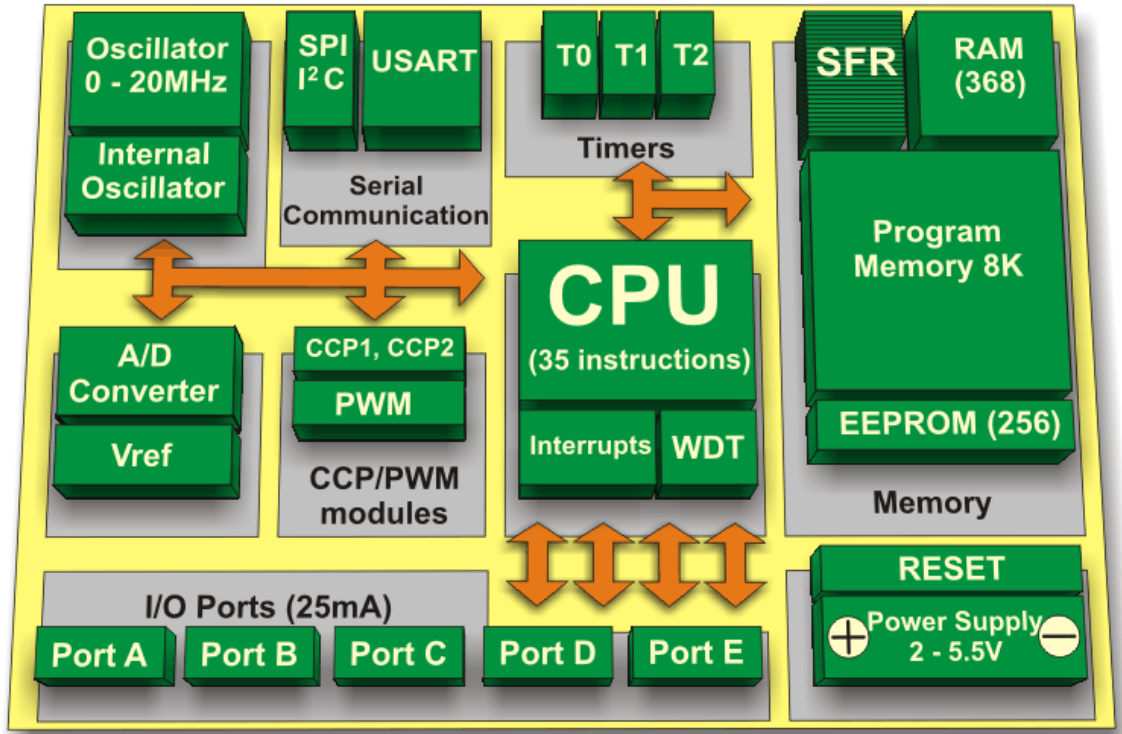
We need clock signals in nearly every electronic circuit: microcontrollers, analog-digital converters (ADCs), communication interfaces like CAN transceivers and USB interfaces etc.

There are different circuit topologies for clock generation, and within the scope of this lecture, we will learn one of the most fundamental clock generation circuits: ring oscillators. There are also Colpitts oscillators, cross-coupled oscillators and Hartley oscillators, however these oscillator types will not be covered within the scope of this lecture.

Since an electronic system might have different clock signals at different frequencies, we can derive clock signals at lower frequencies by using frequency dividers

SOME APPLICATION EXAMPLES

Microcontrollers, Digital Signal Processors etc.



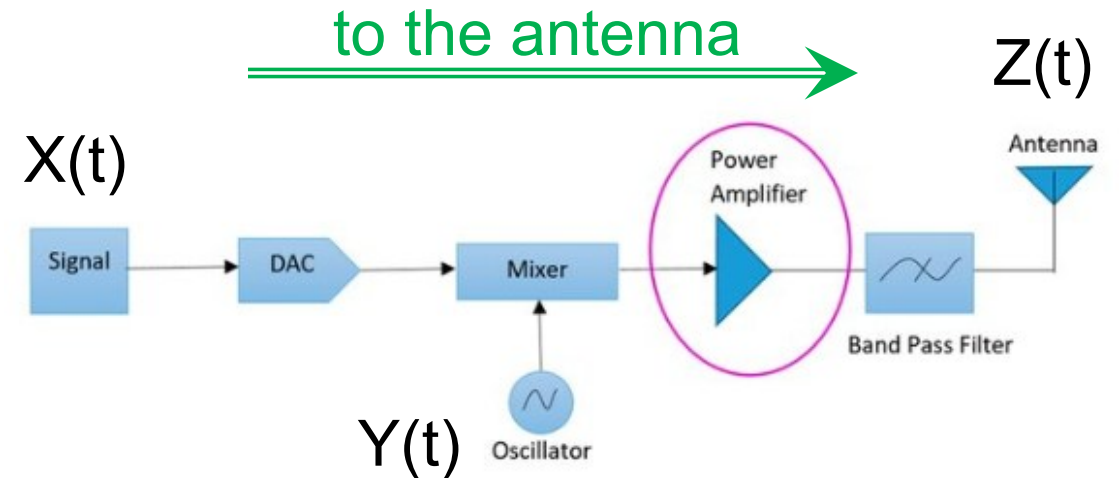
PIC16F microcontroller with an internal oscillator. Oscillator signal used in CPU, ADC, Timers, comm interfaces etc.

Communication Circuits

$X(t)$: low frequency data signal $\rightarrow$ generated by e.g., a digital signal processor

$Y(t)$: clock signal

$Z(t)$: high frequency transmit signal at the antenna



Oscillator signal used in radio frequency transmitter

$$X(t) = A \cdot \cos(2\pi f_1 t)$$

$$Y(t) = B \cdot \cos(2\pi f_2 t)$$

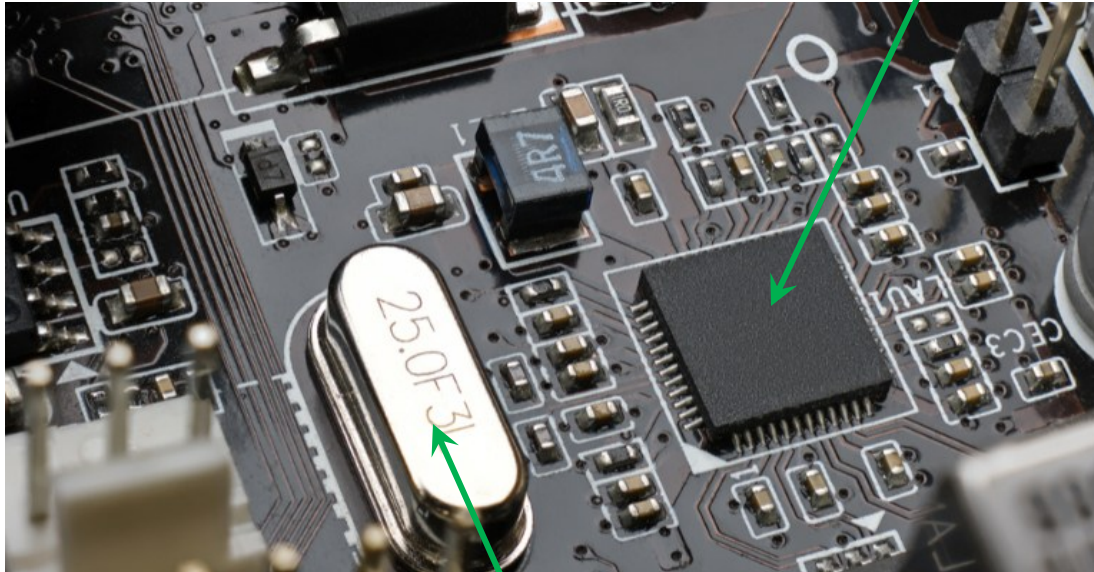
$$Y(t) = \frac{AB}{2} \cdot [\cos(2\pi(f_1 + f_2)t) + \cos(2\pi(f_1 - f_2)t)]$$

High Frequency Term at the Antenna

Suppressed by the Band Pass Filter

SOME APPLICATION EXAMPLES

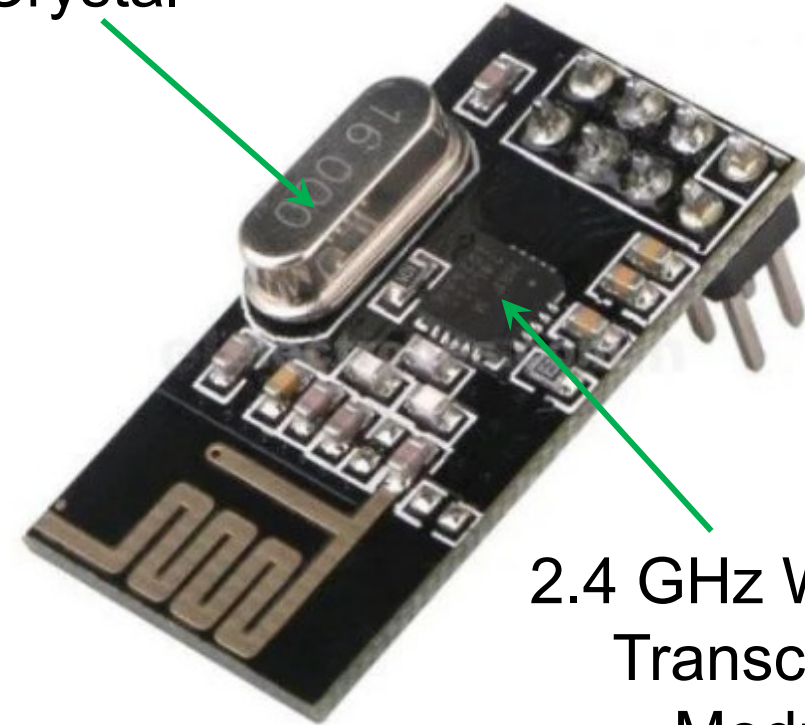
Microcontrollers, Digital Signal Processors etc.



Crystal

Communication Circuits

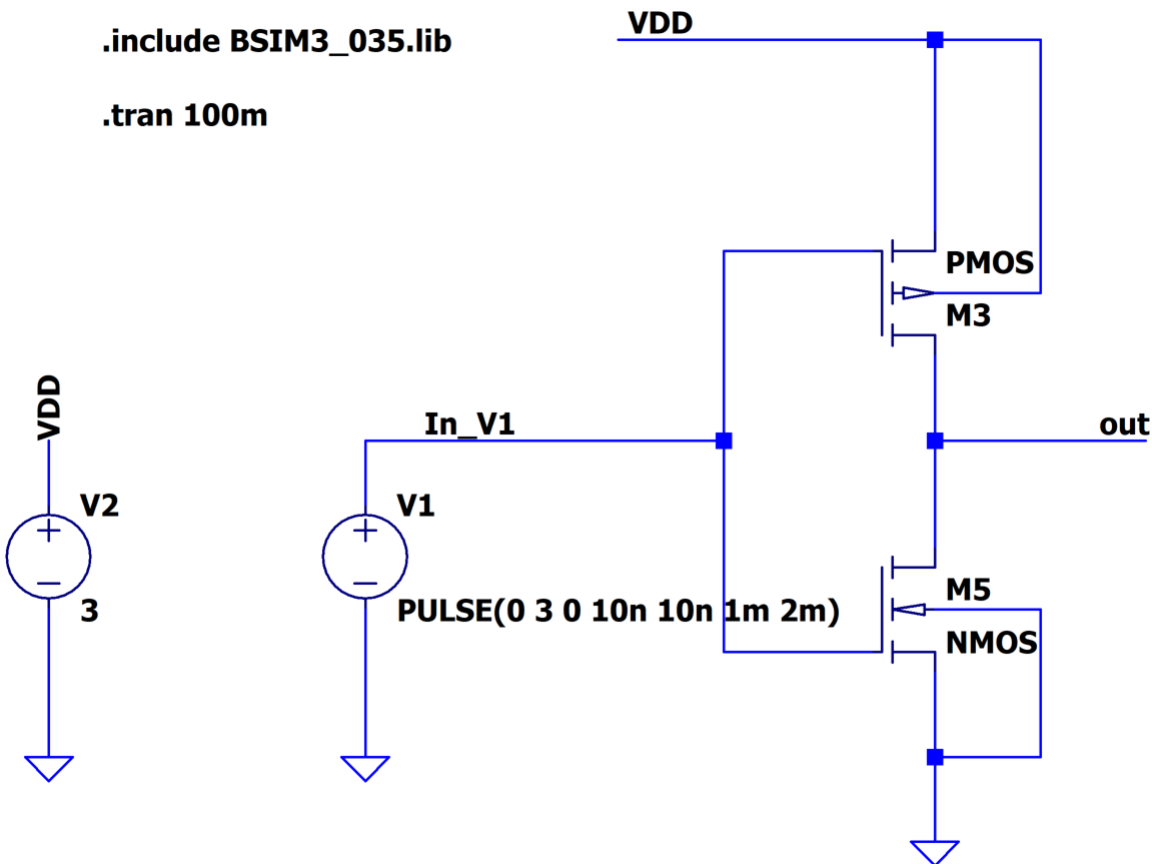
Crystal



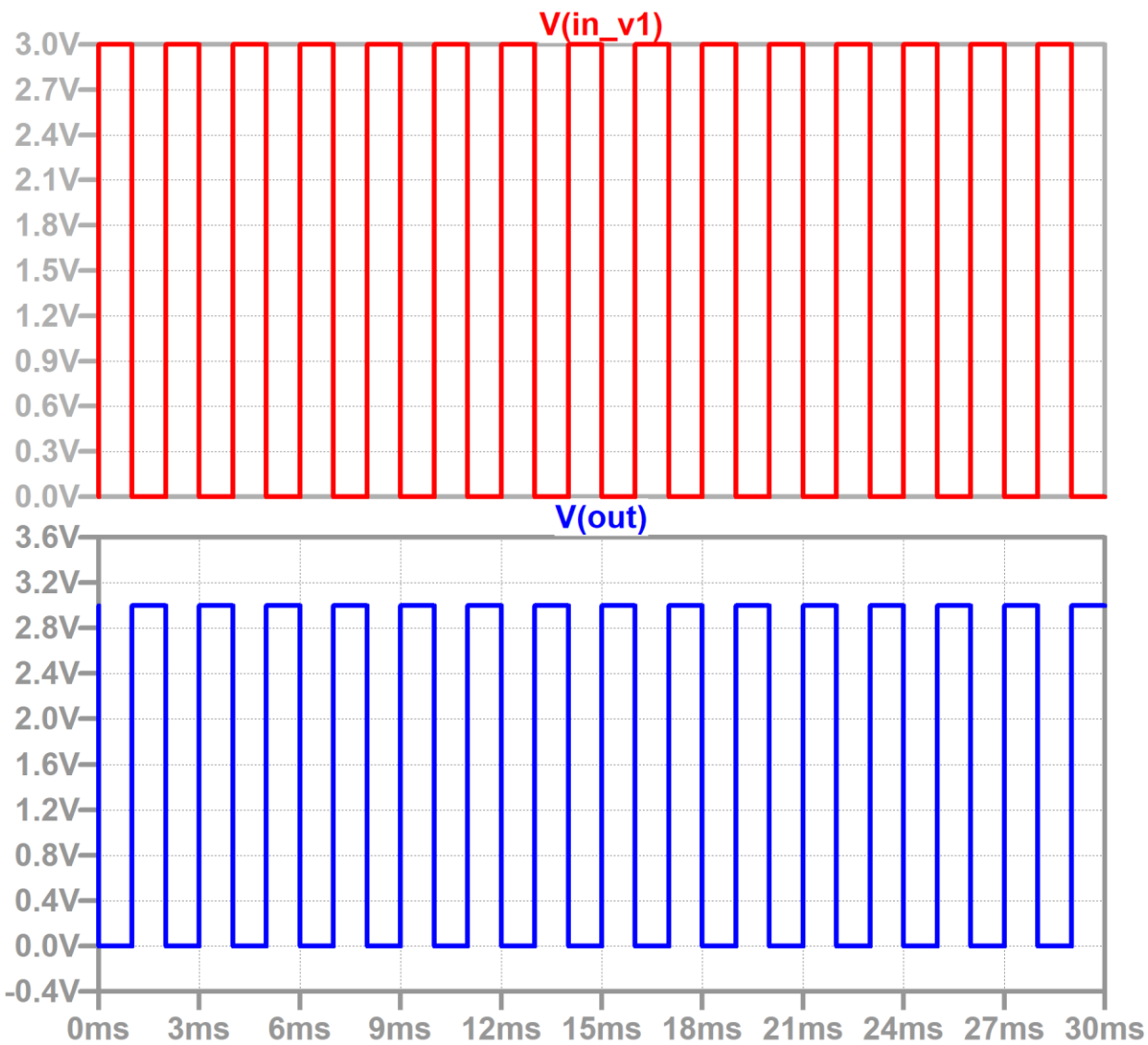
2.4 GHz Wireless
Transceiver
Module

Crystals will be covered in the next slides!

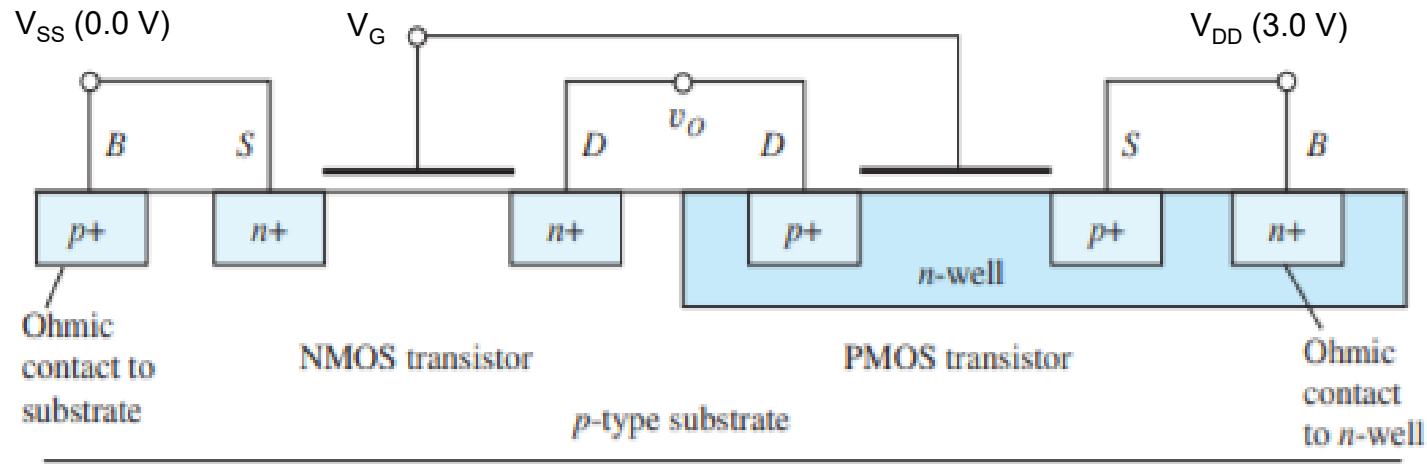
CMOS INVERTERS



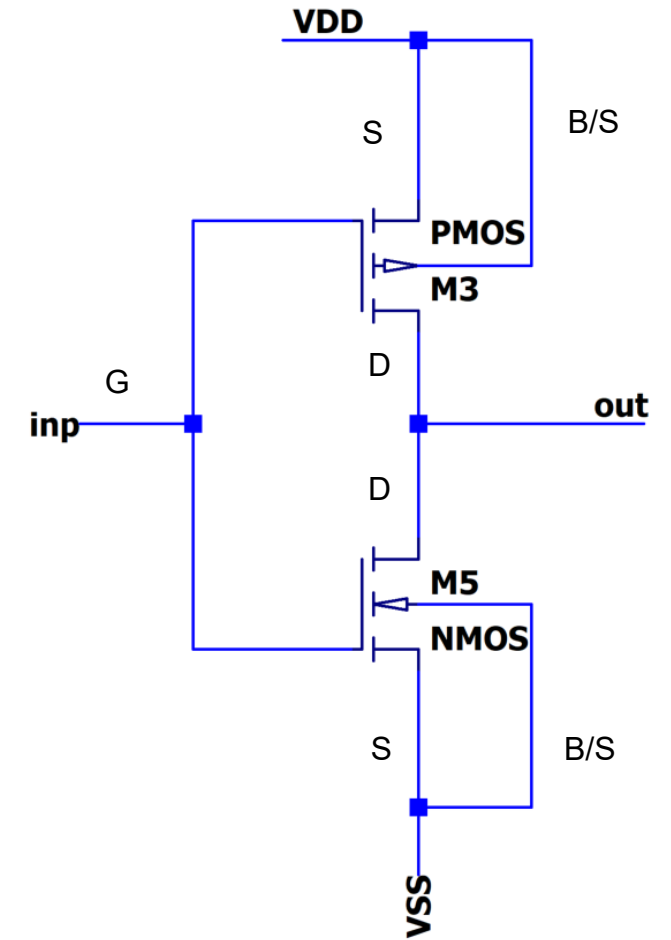
inverter_sim.asc



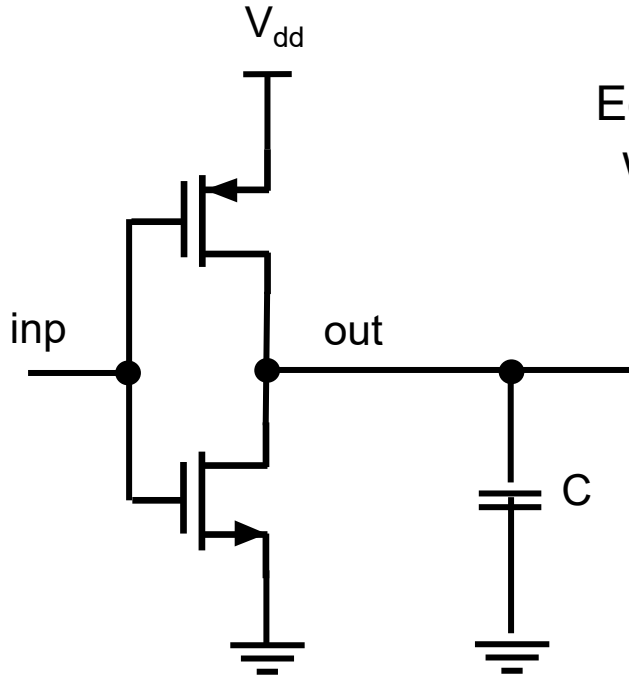
CMOS INVERTERS



D: Drain
G: Gate
S: Source
B: Bulk/Substrate

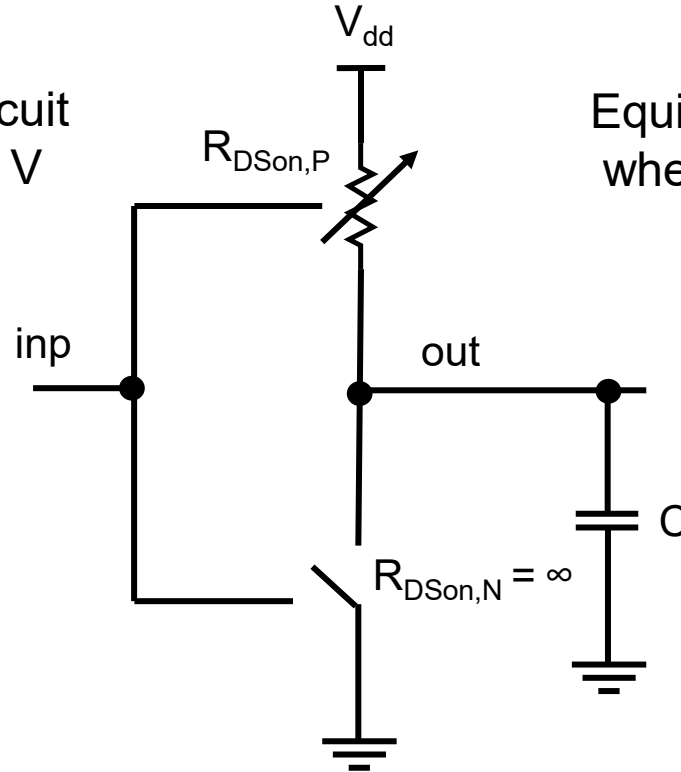


CMOS INVERTERS



Equivalent Circuit
when $inp = 0\text{ V}$

*** $out = V_{DD} \left(1 - e^{-\frac{t}{R_{DSon,P} \cdot C}} \right)$

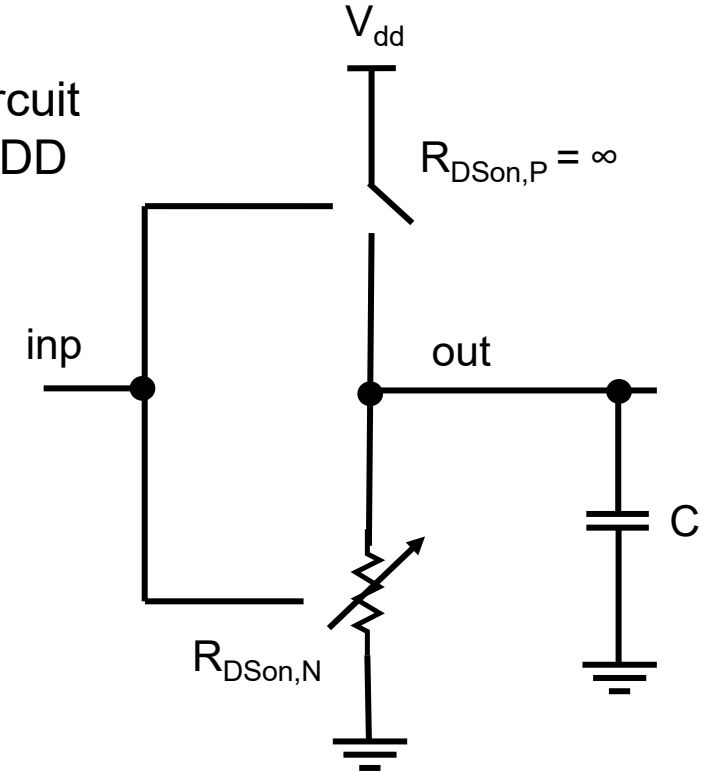


pmos conducts
and nmos does
not conduct

Equivalent Circuit
when $inp = V_{DD}$

C is charged by V_{DD} via $R_{DSon,P}$

*** $out = V_{DD} \left(e^{-\frac{t}{R_{DSon,N} \cdot C}} \right)$



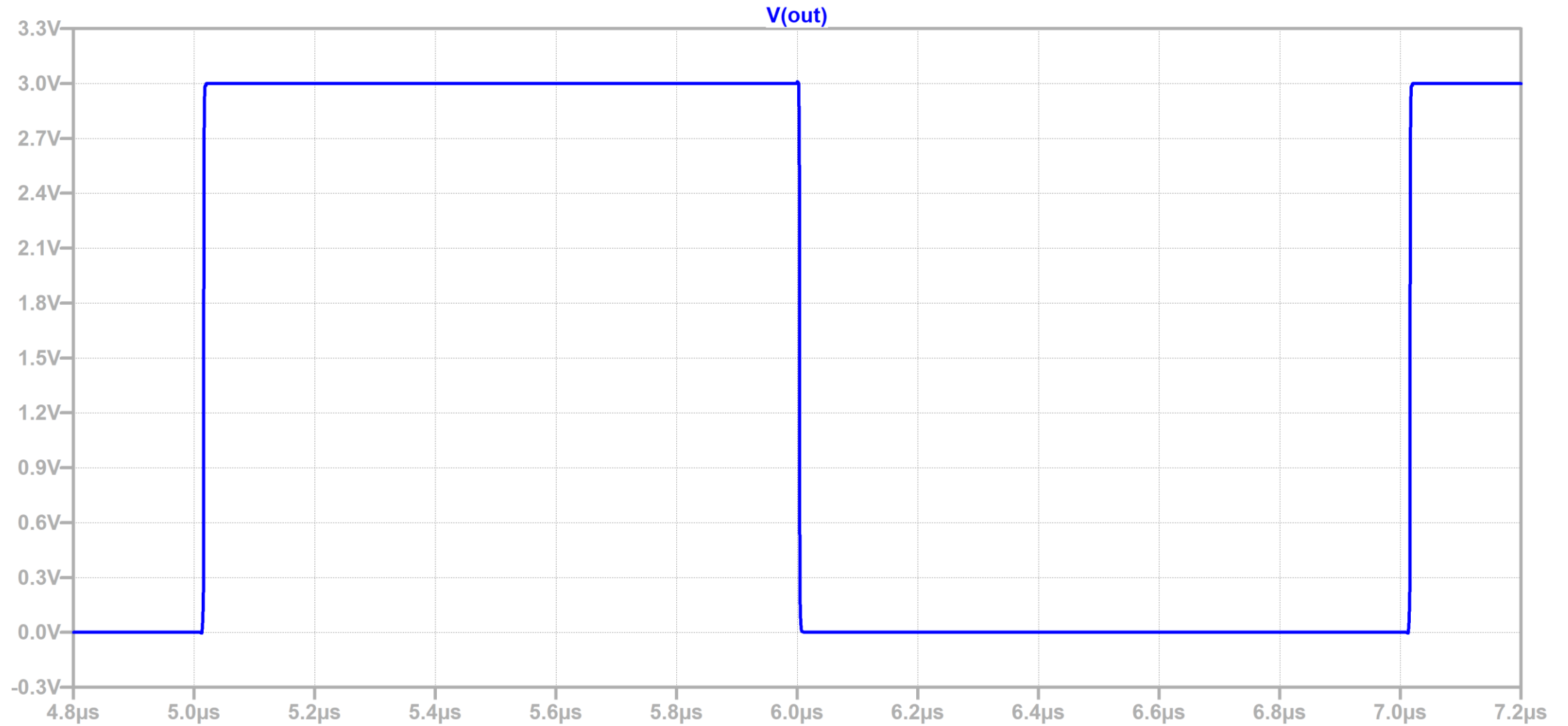
pmos conducts
and nmos does
not conduct

$R_{DSon,P}$ and $R_{DSon,N}$ values depend on V_{GS}

Although the transistor spends a significant portion of the transition in saturation, the oversimplified equations incorrectly assume a variable-resistor model, which is strictly valid only in the linear region.

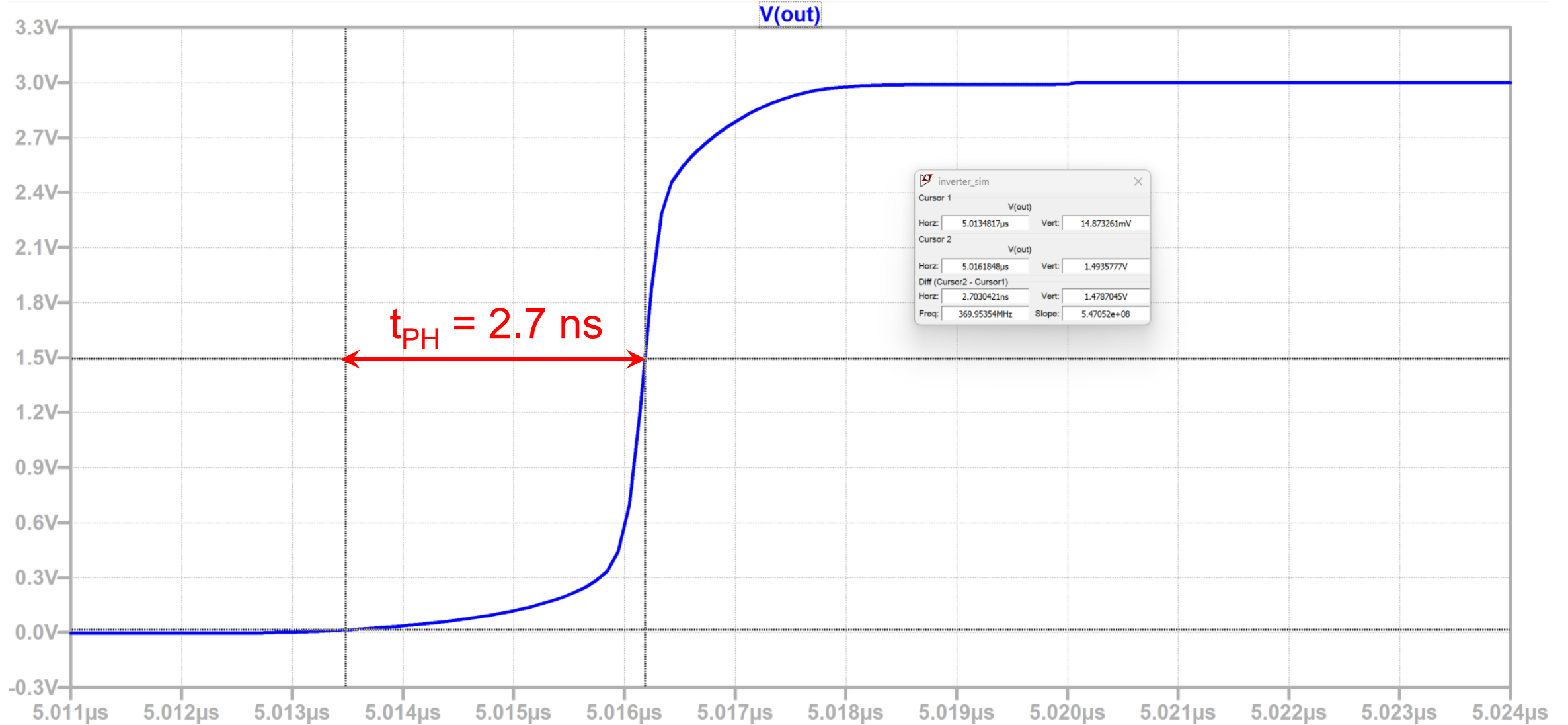
CMOS INVERTERS

Propagation Delay of CMOS Inverters



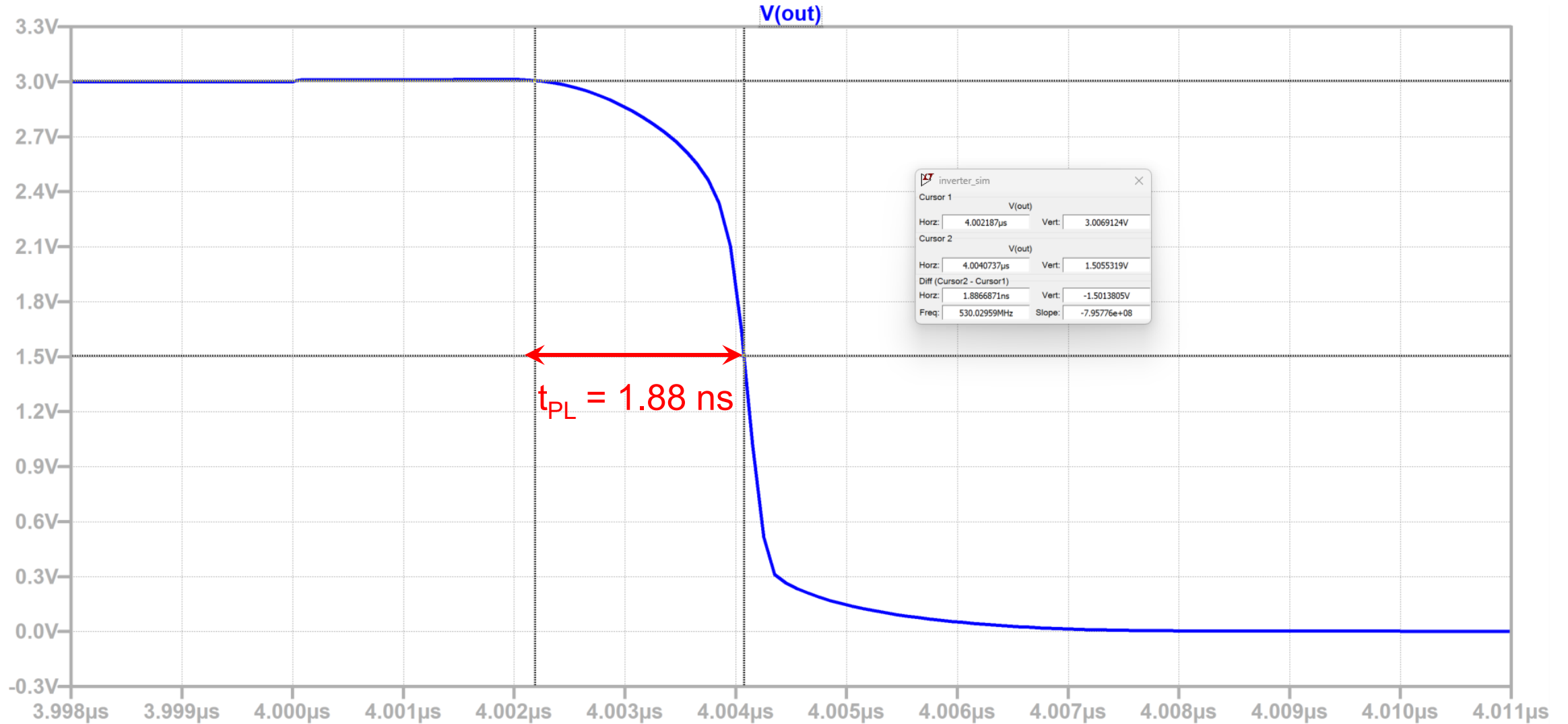
CMOS INVERTERS

Propagation Delay of CMOS Inverters



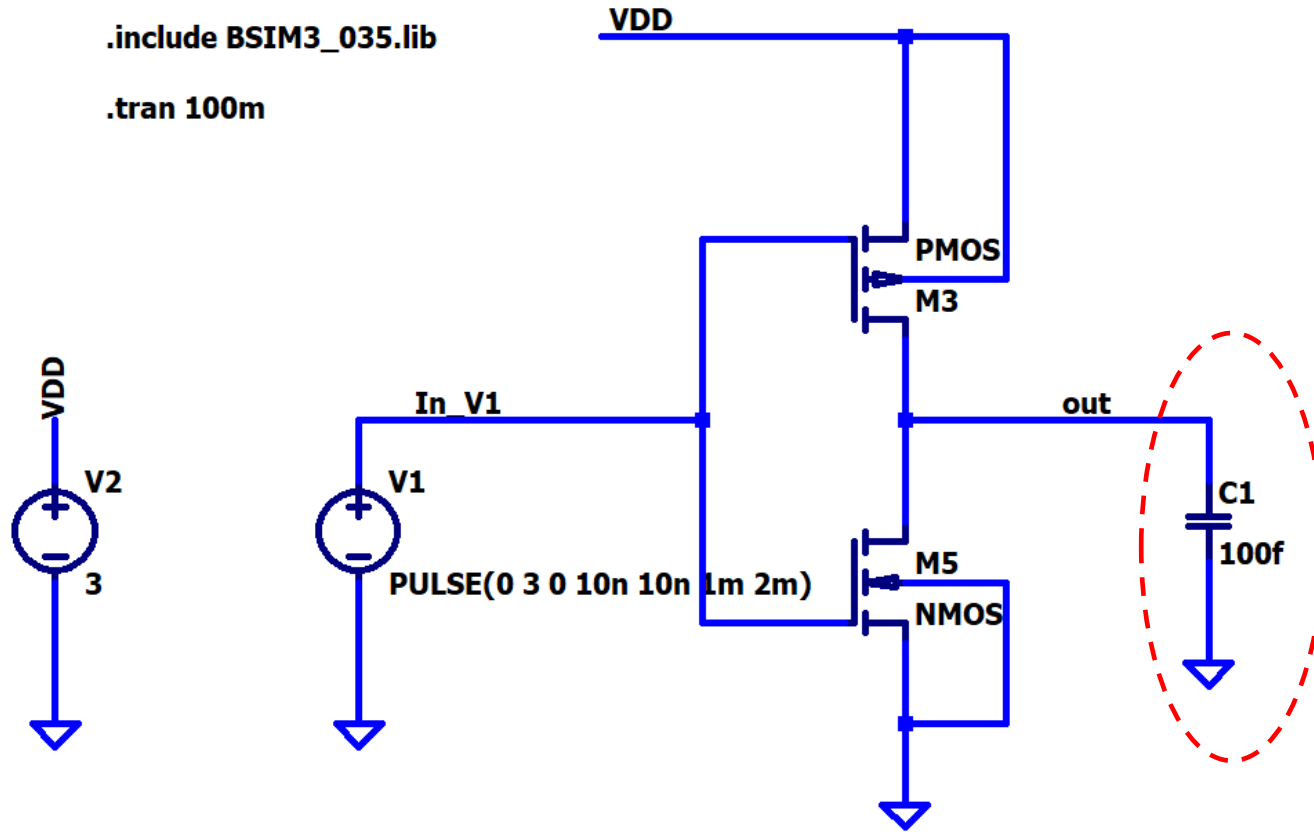
CMOS INVERTERS

Propagation Delay of CMOS Inverters $\rightarrow t_p = (t_{PH} + t_{PL})/2 = 2.29 \text{ ns}$



CMOS INVERTERS

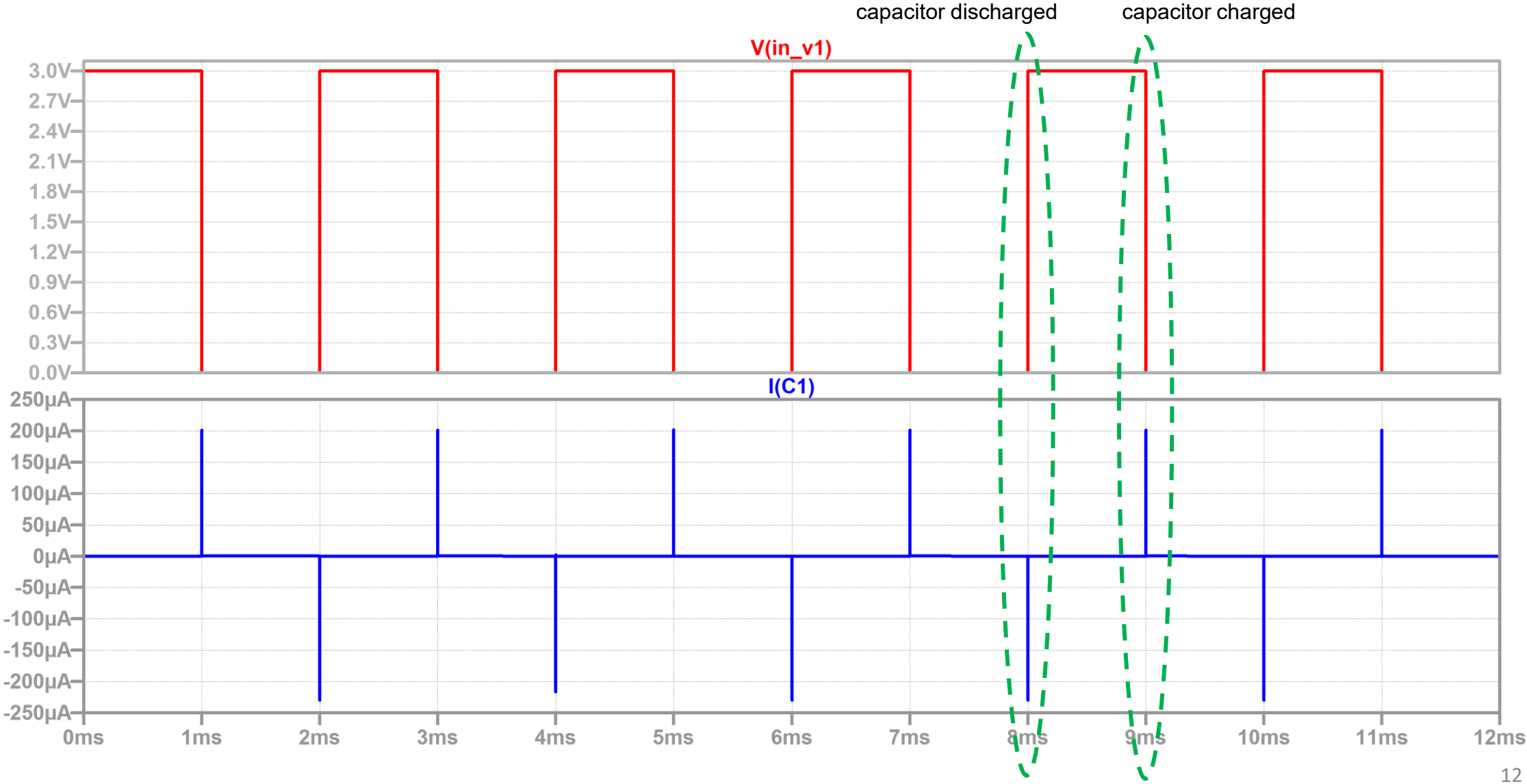
CMOS Inverter with Capacitive Load



If two CMOS inverters are cascaded (output of one inverter is connected to the input of another inverter), then this capacitor might represent/model the input capacitance of the next CMOS inverter stage!

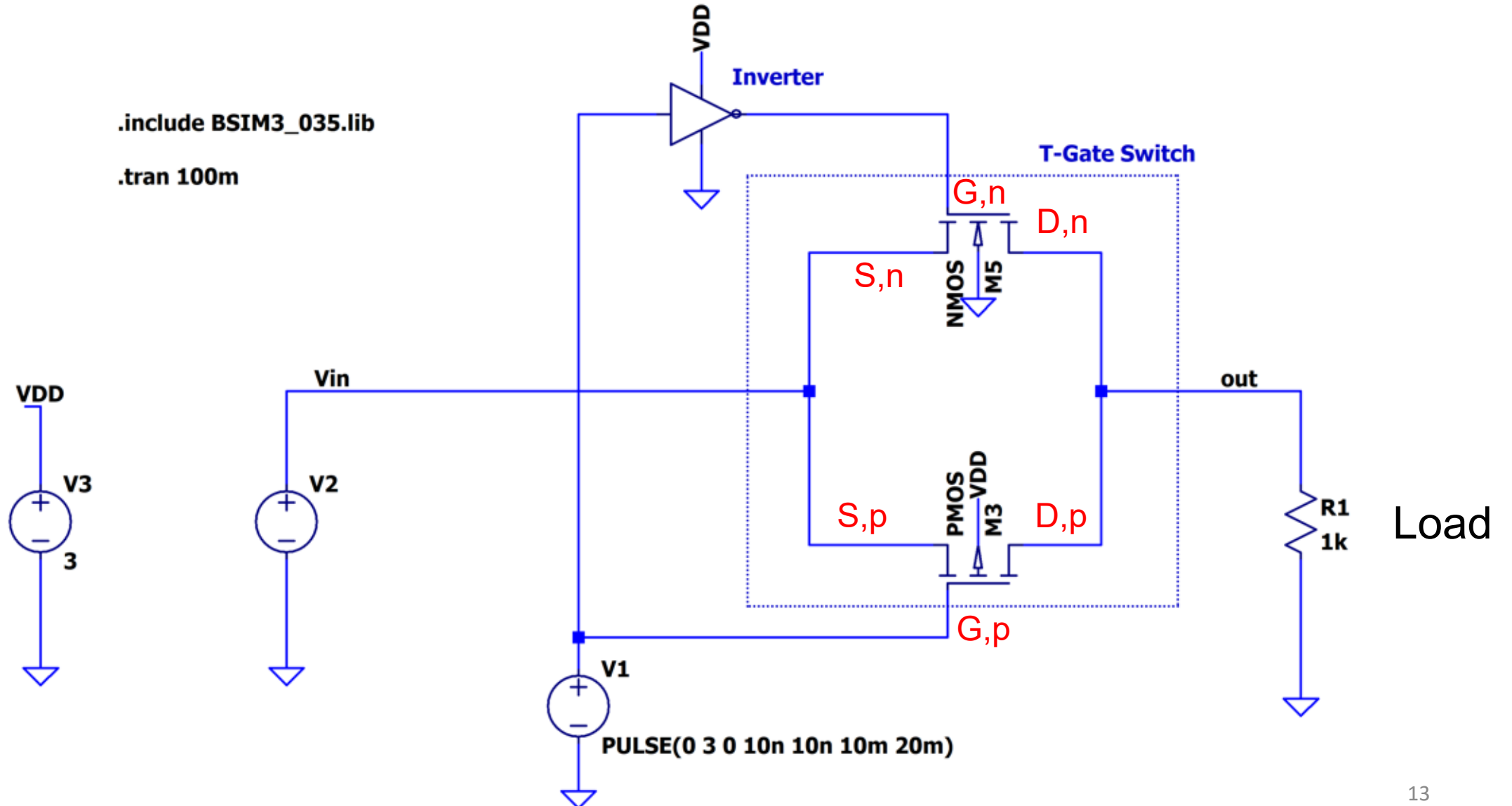
CMOS INVERTERS

CMOS Inverter with Capacitive Load



TRANSMISSION GATES (T GATES)

Transmission gate acts as tristate buffer/switch with only two transistors:



TRANSMISSION GATES (T GATES)

Why do we use pmos and nmos at the same time?

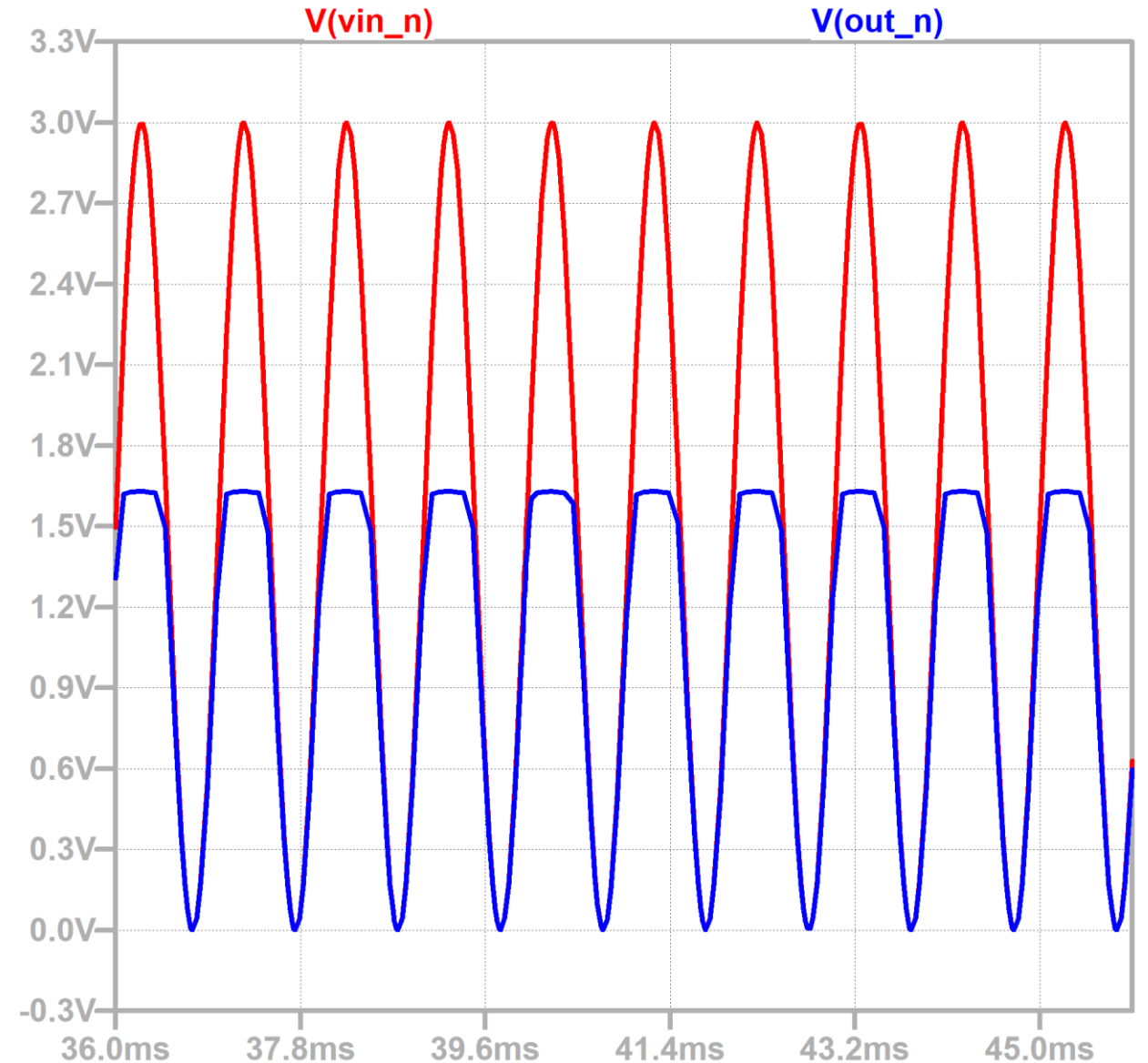
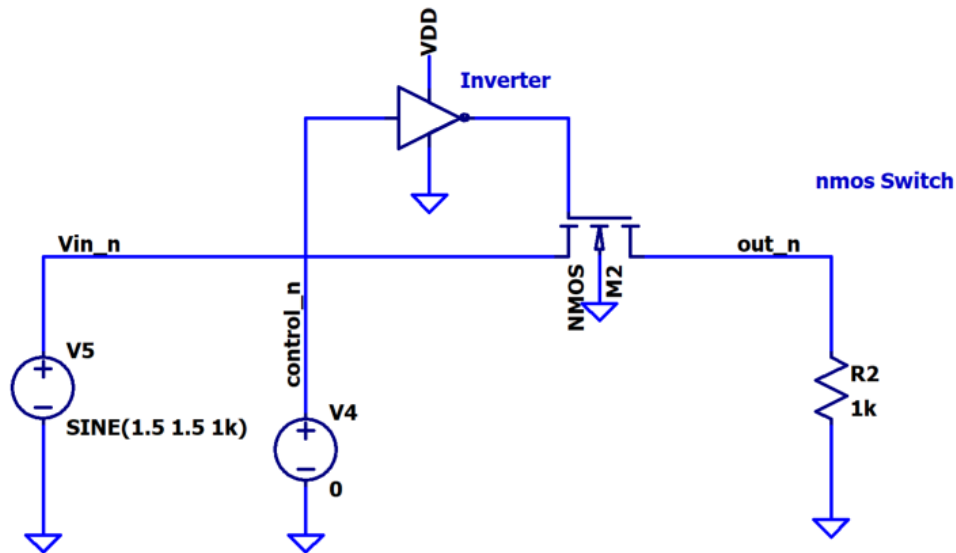
Example:

We remove the pmos transistor and we use only the nmos as a switch

If, for example, $V_{in} = 3\text{ V}$, $V_{GS,n}$ becomes 0 V. $V_{GS,n} = 0\text{ V} < V_{GSn,th}$, nmos cannot be turned on!!!

If we have both pmos and nmos and $V_{in} = 0\text{ V}$, nmos will not conduct but pmos will conduct, since

$$-V_{GS,p} > -V_{GS,p,th}$$



TRANSMISSION GATES (T GATES)

Why do we use pmos and nmos at the same time?

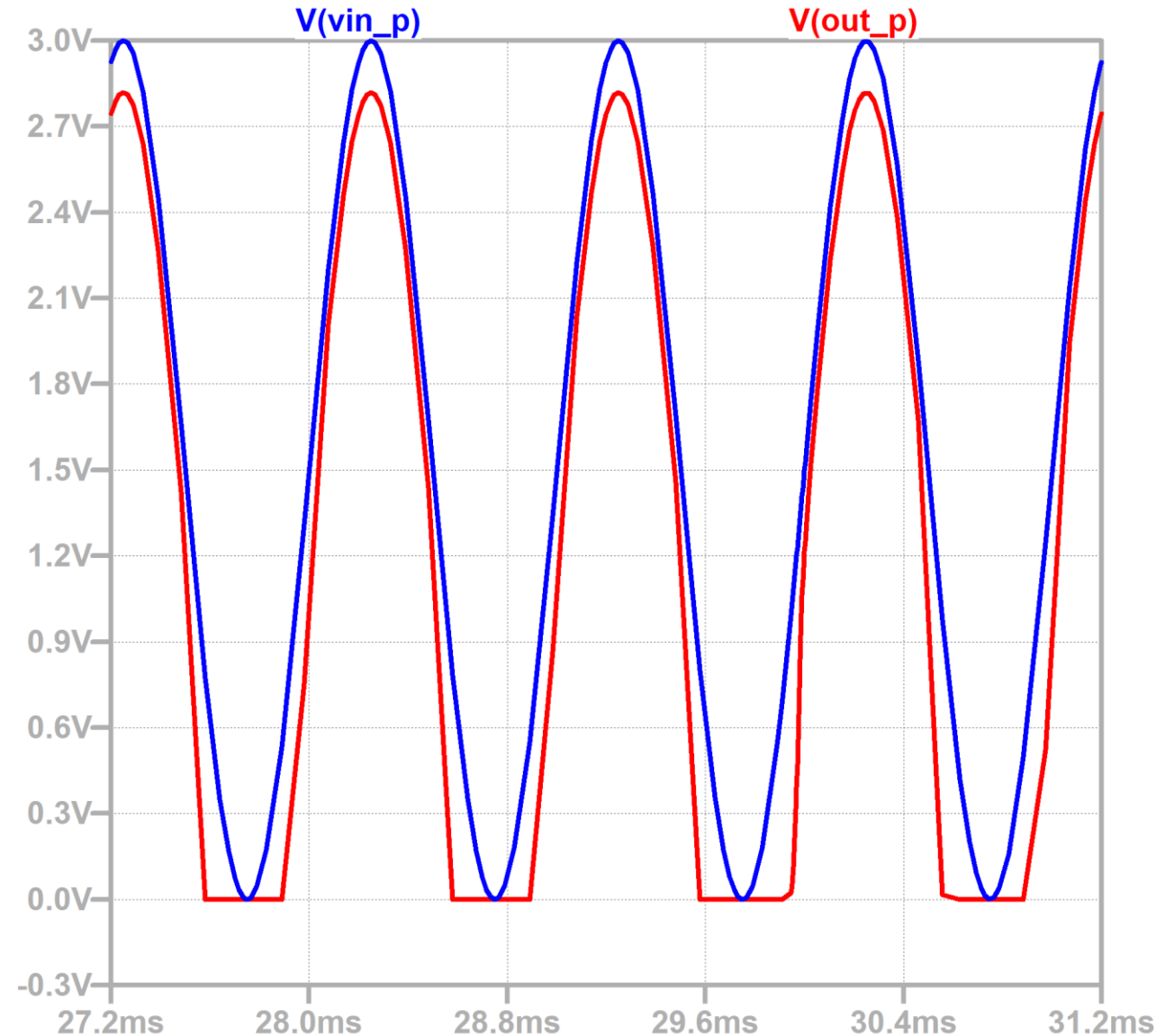
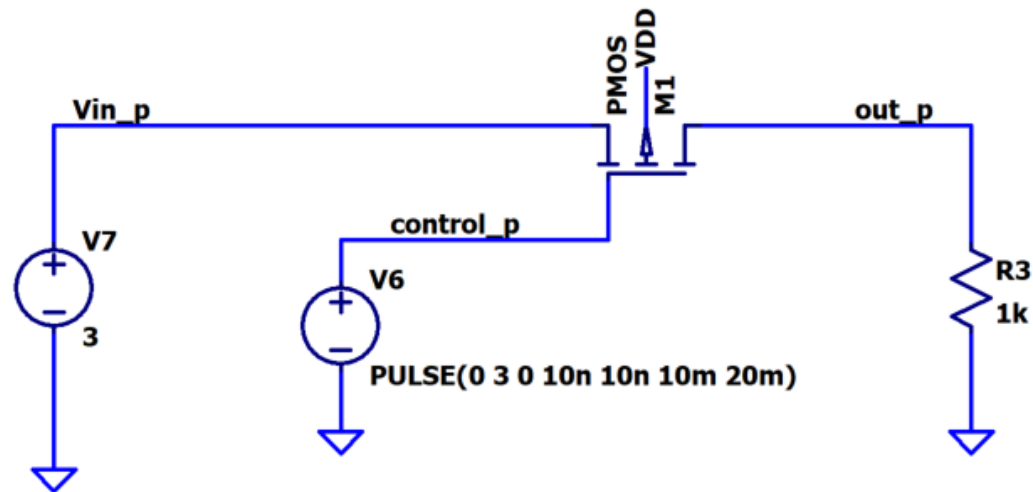
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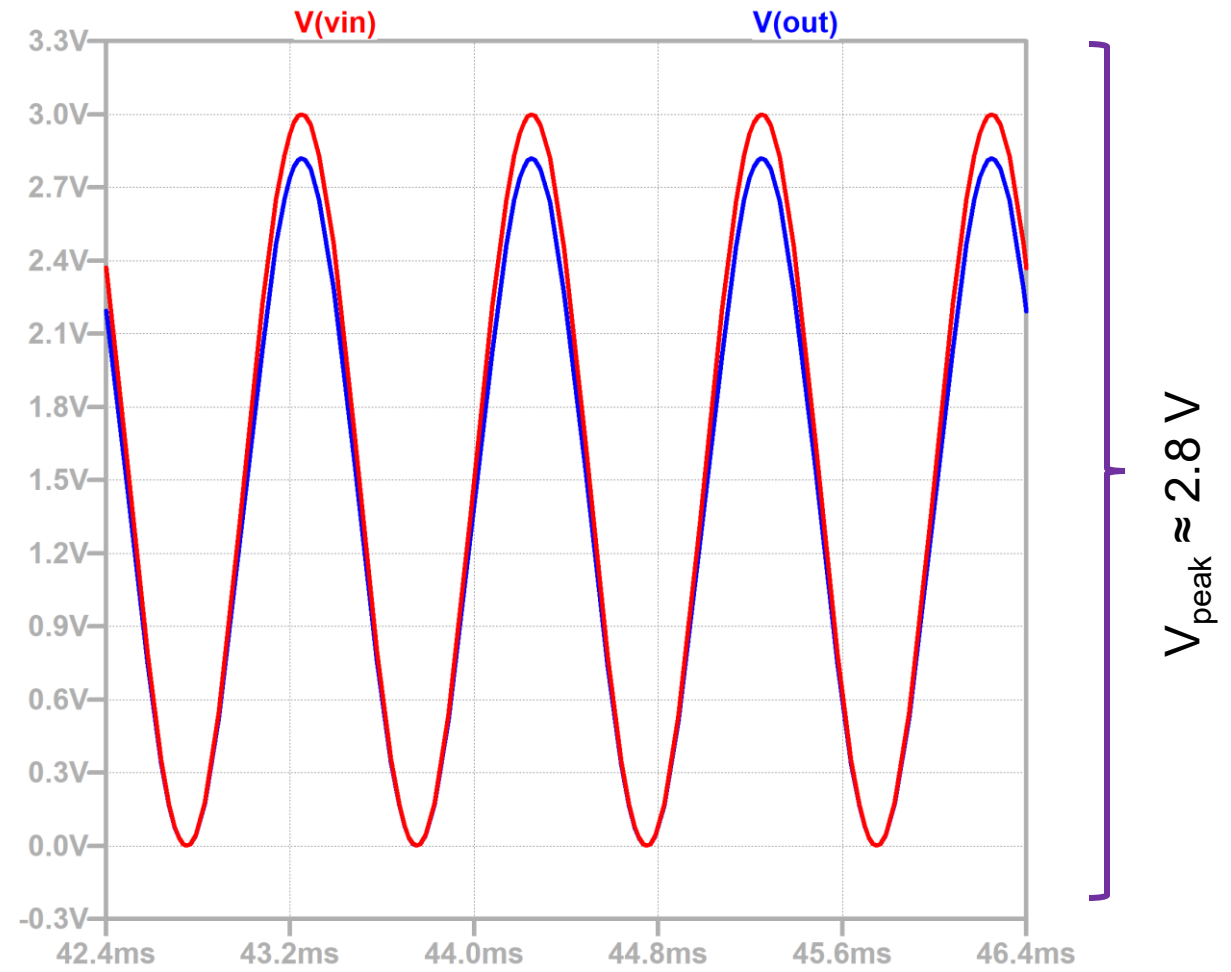
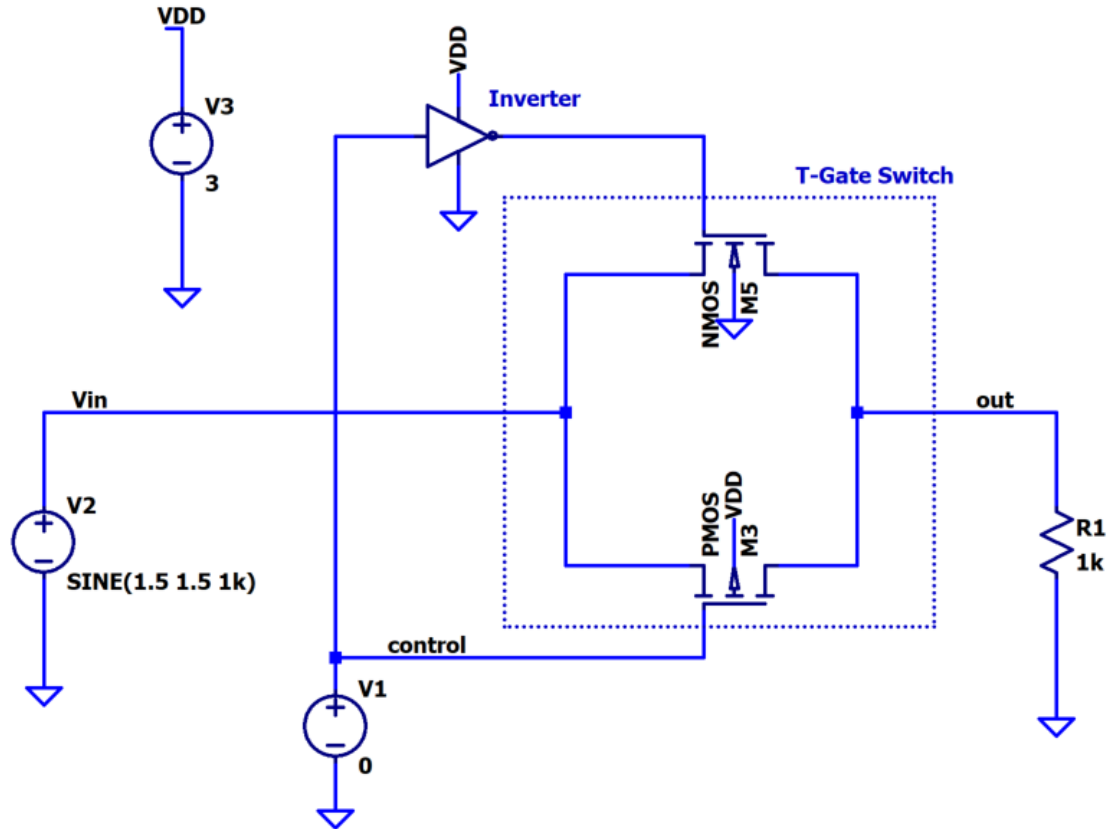
If $V_{in} = 0\text{ V}$, $V_{GS,n}$ becomes 0 V . $V_{GS,n} = 0\text{ V} < V_{GSn,th}$, nmos cannot be turned on!!!

If we have both pmos and nmos and $V_{in} = 0\text{ V}$, nmos will not conduct but pmos will conduct, since

$$-V_{GS,p} > -V_{GSp,th}$$

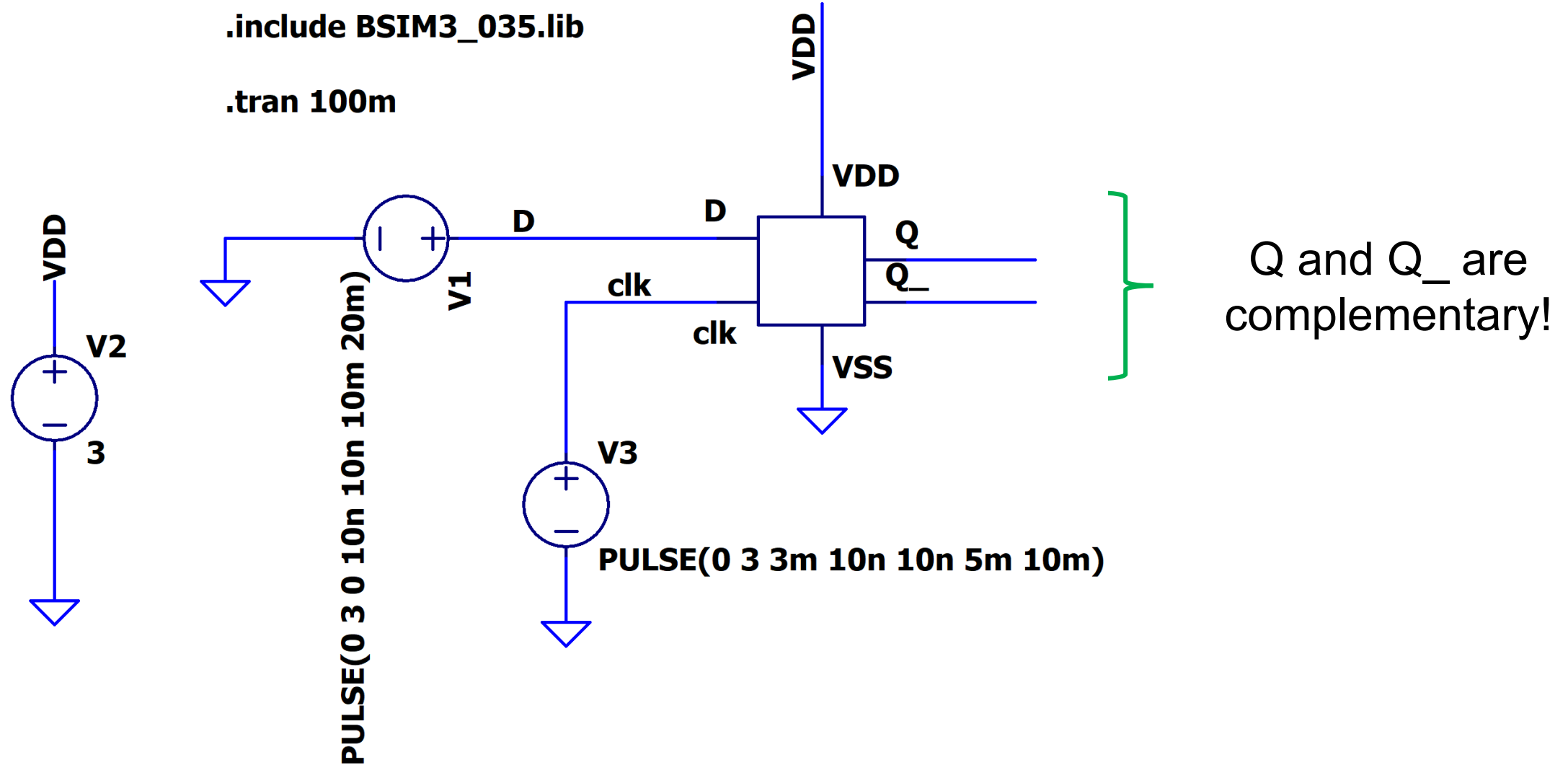


TRANSMISSION GATES (T GATES)

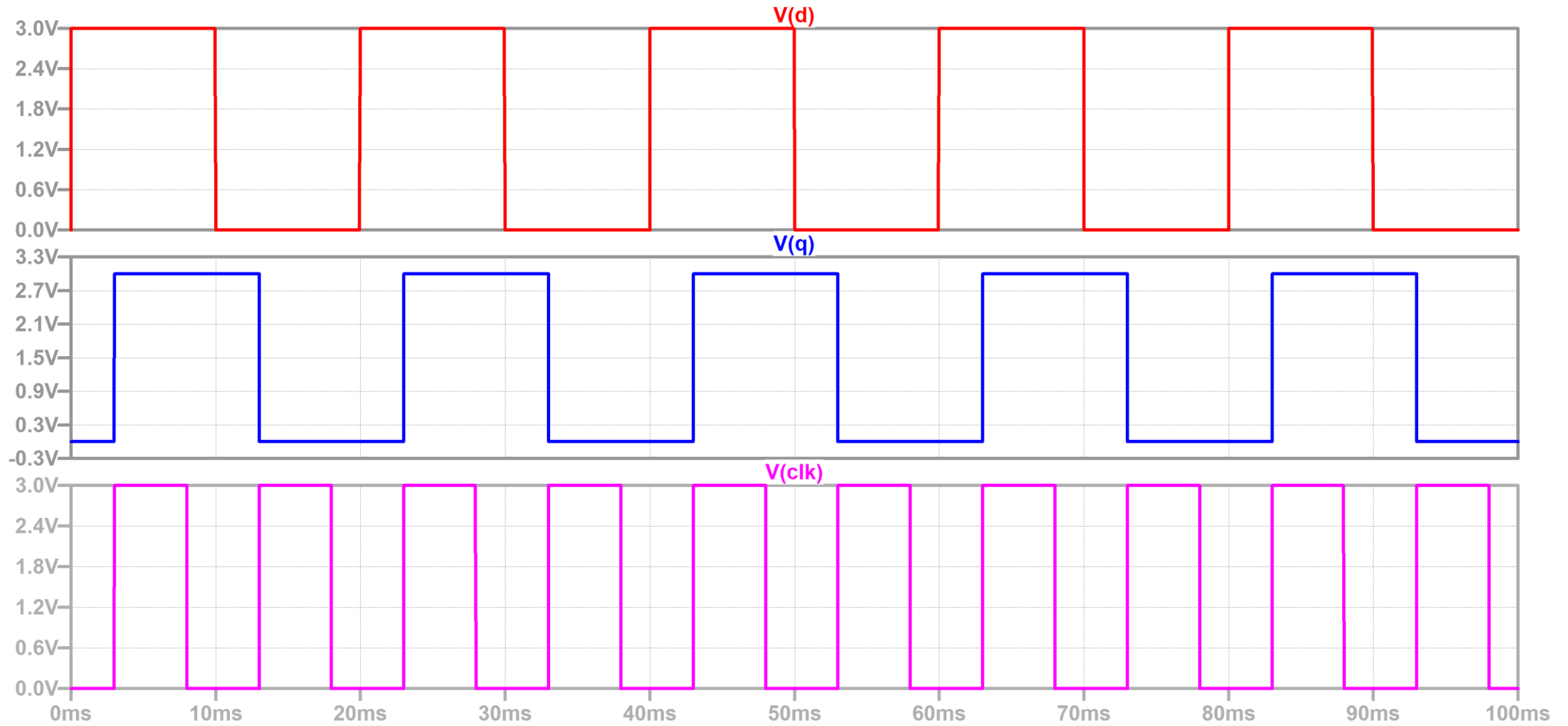


Transmission Gate works as a switch for an input voltage range of $V_{in} = 0 \dots 3V$ without any problems!

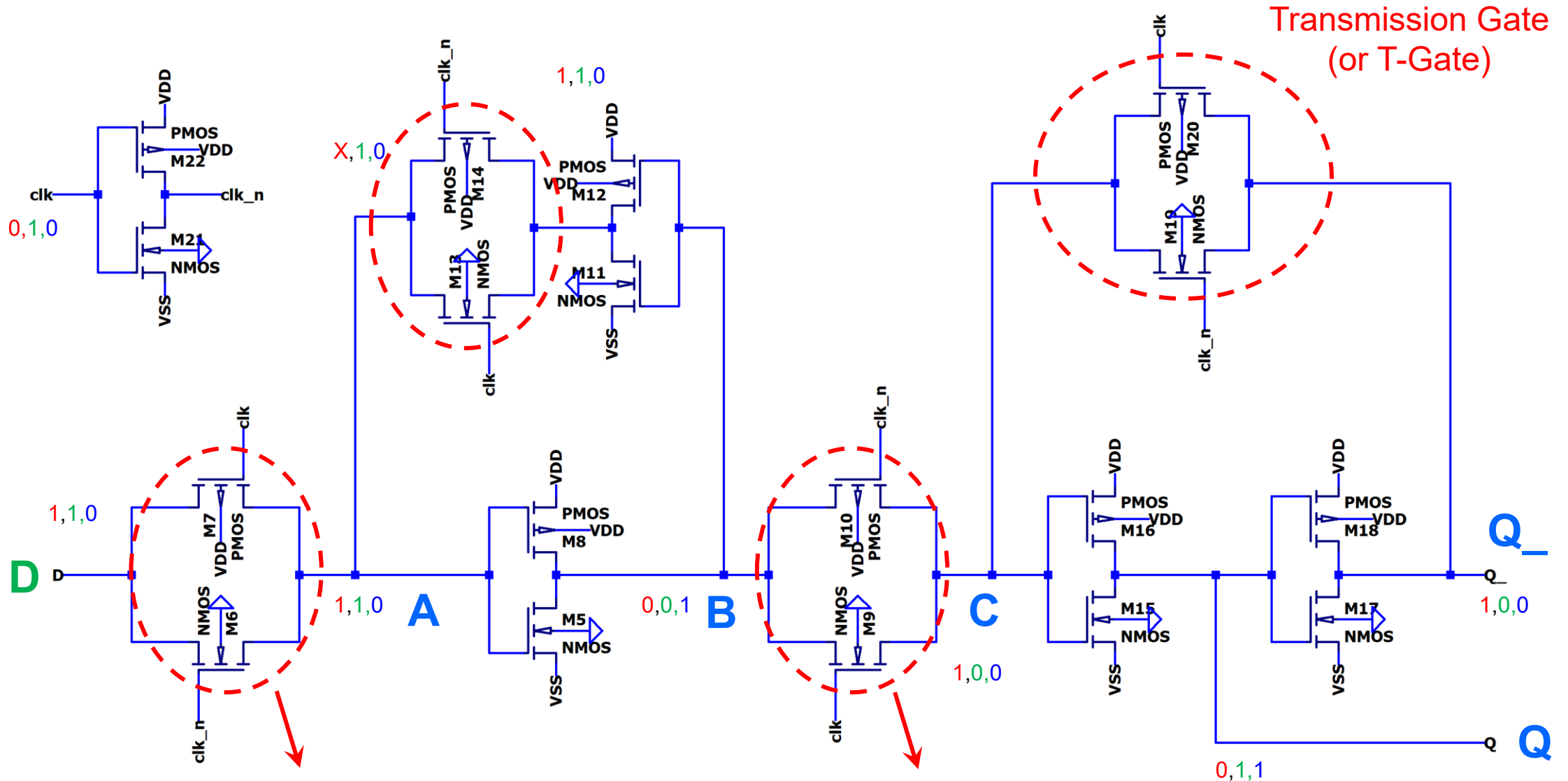
CMOS POSITIVE EDGE TRIGGERED D-TYPE FLIP FLOP



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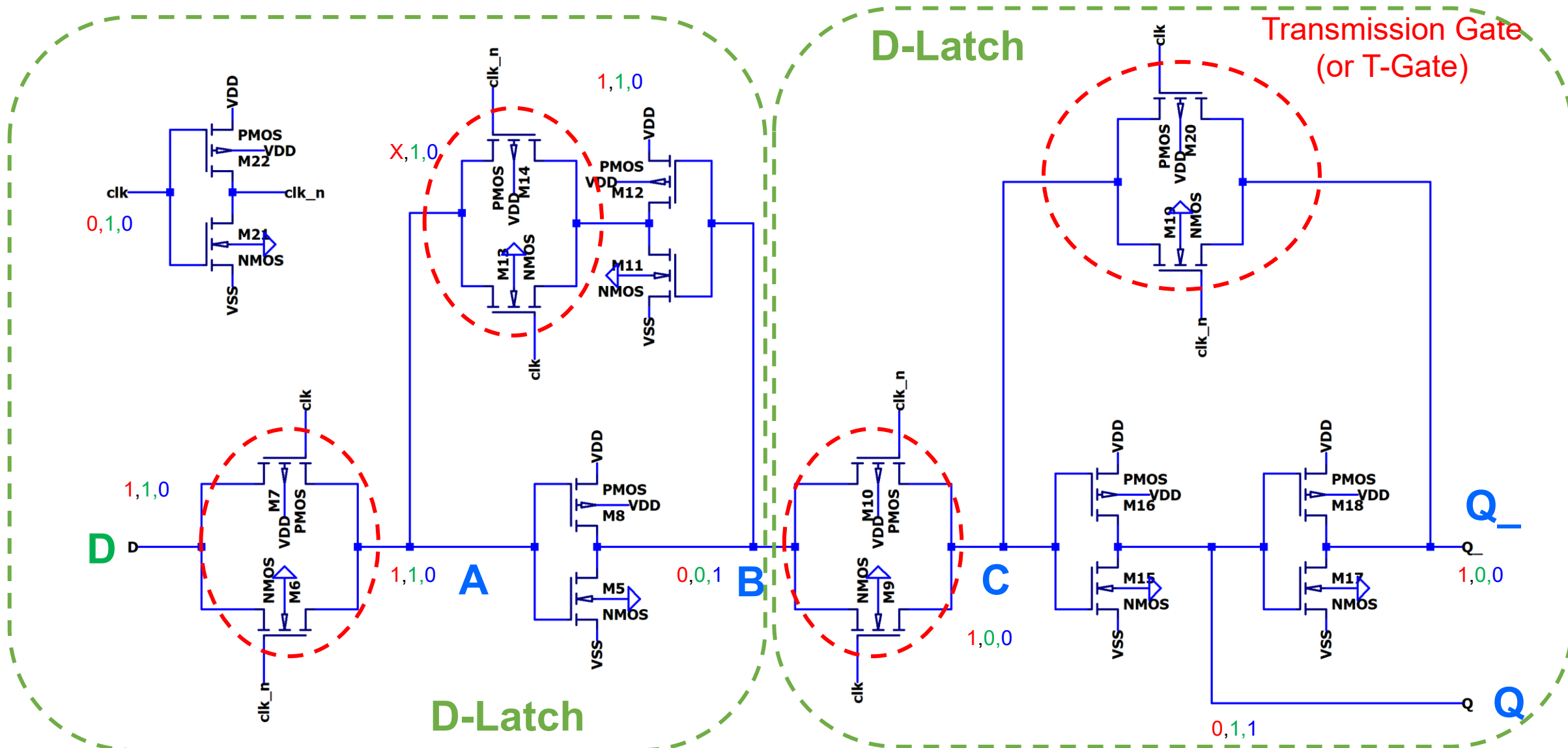


As long as $\text{clk}=1$, this T-Gate will not let D to be shifted from D to A (T-Gate open) $\rightarrow A = B'$ & no change

As long as $\text{clk}=0$, this T-Gate will not let B to be shifted to C (T-Gate open) $\rightarrow C = Q_ \&$ no change

An initial condition of 0 is assumed at Q

CMOS POSITIVE EDGE TRIGGERED D-TYPE FLIP FLOP

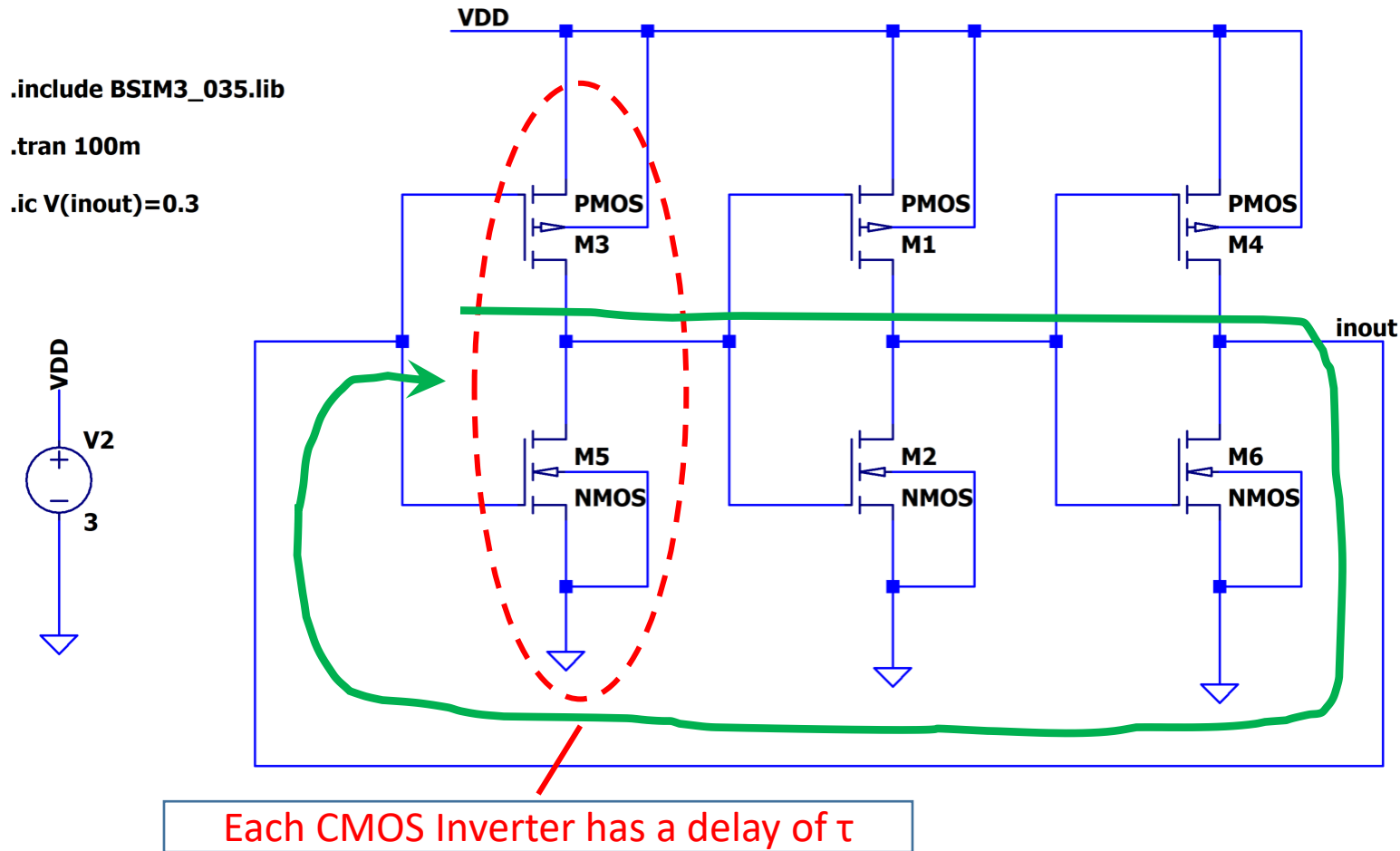


If clk transitions from 0 to 1, C becomes equal to B, therefore, Q becomes equal to D!

Two D-latches form a D type flip flop

CMOS RING OSCILLATORS

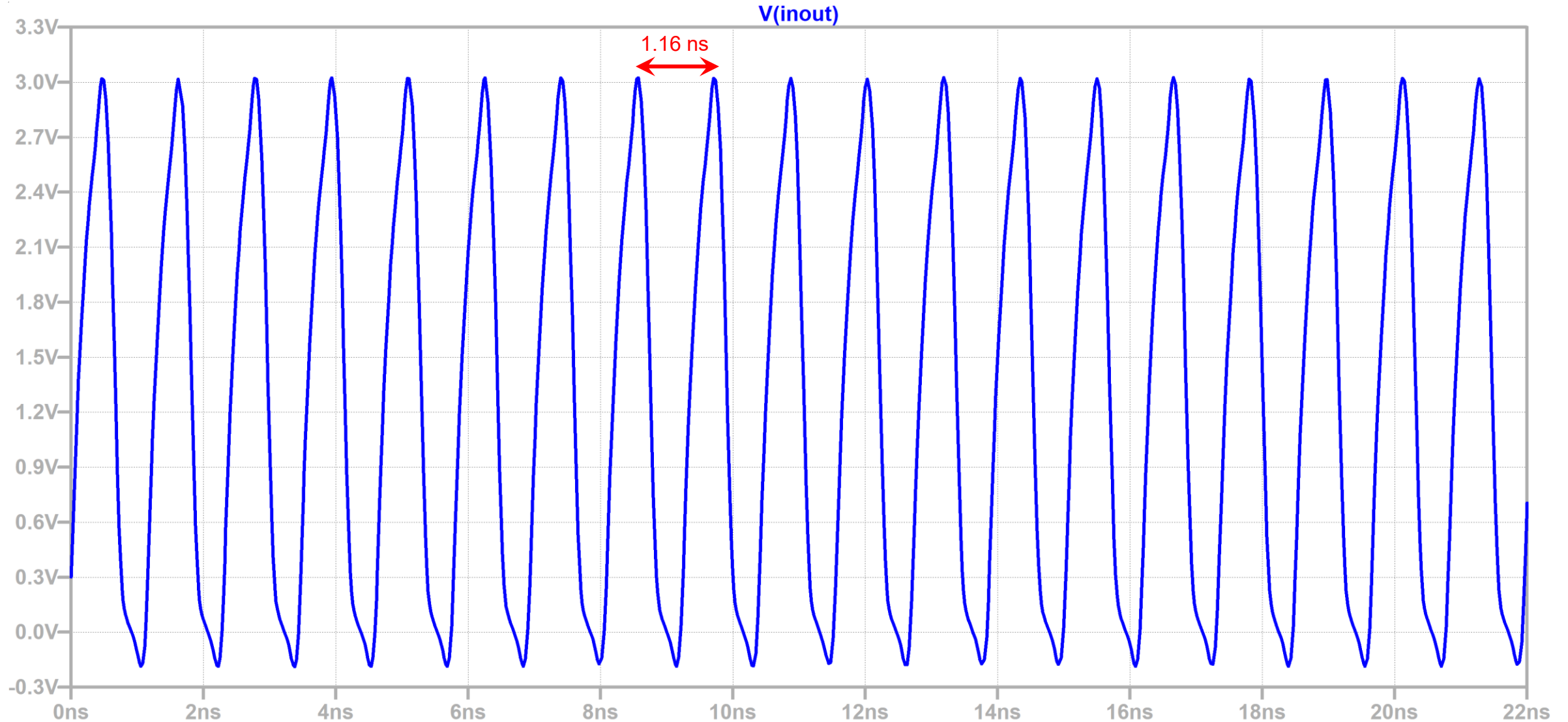
Circuit has no input!



If the initial condition of net “inout” is e.g., 0 V (or 5 V or 0.3 V...), then after 3τ , “inout” becomes 3 V. After another 3τ , “inout” becomes 0 V again. Therefore, the output signal becomes an oscillating signal with a period of 6τ . The period can be changed by changing the number of inverters or placing a capacitors at the output of each inverter!

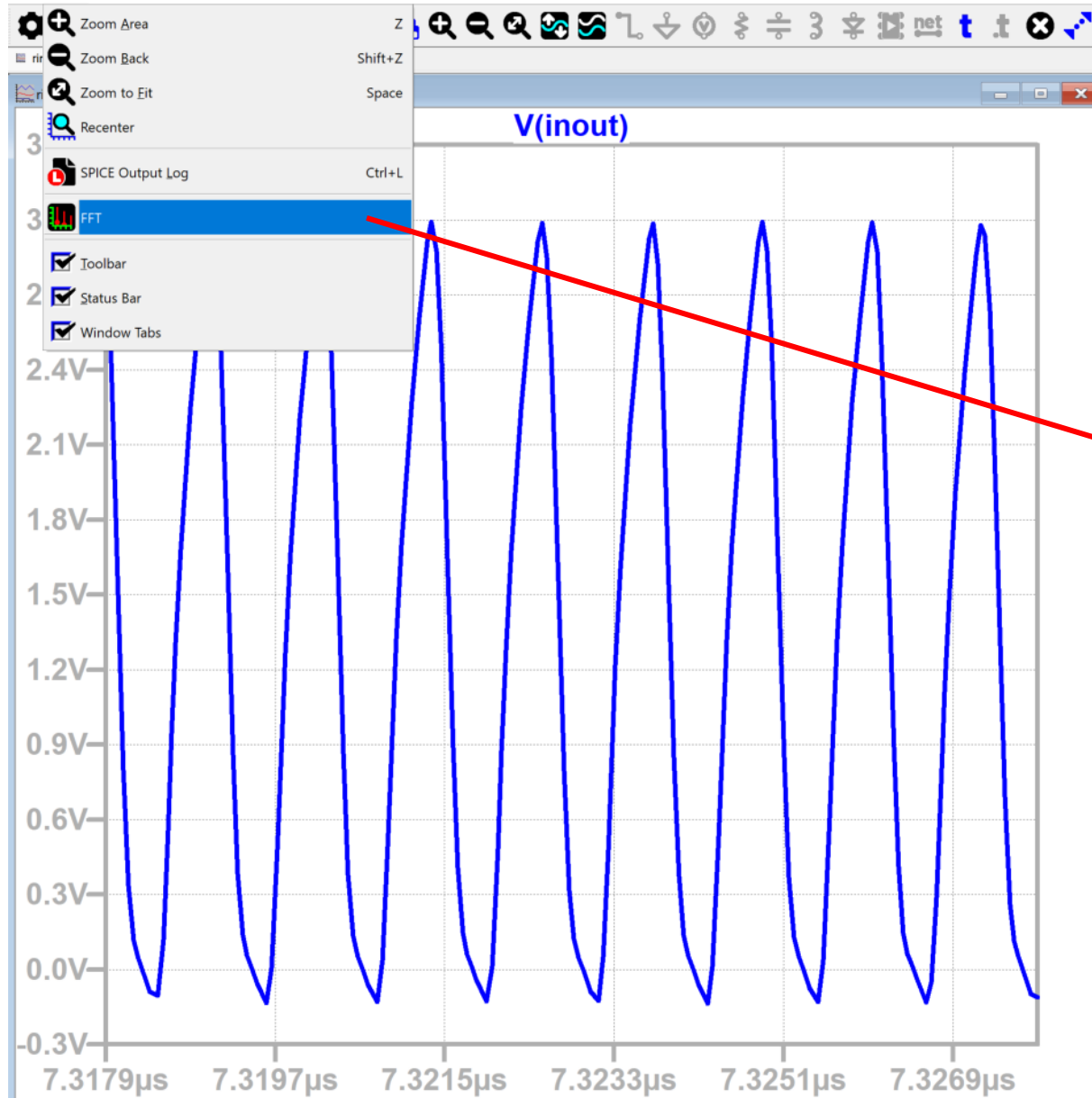
CMOS RING OSCILLATORS

Period = 1.16 ns $\rightarrow$ Frequency $\approx$ 862 MHz



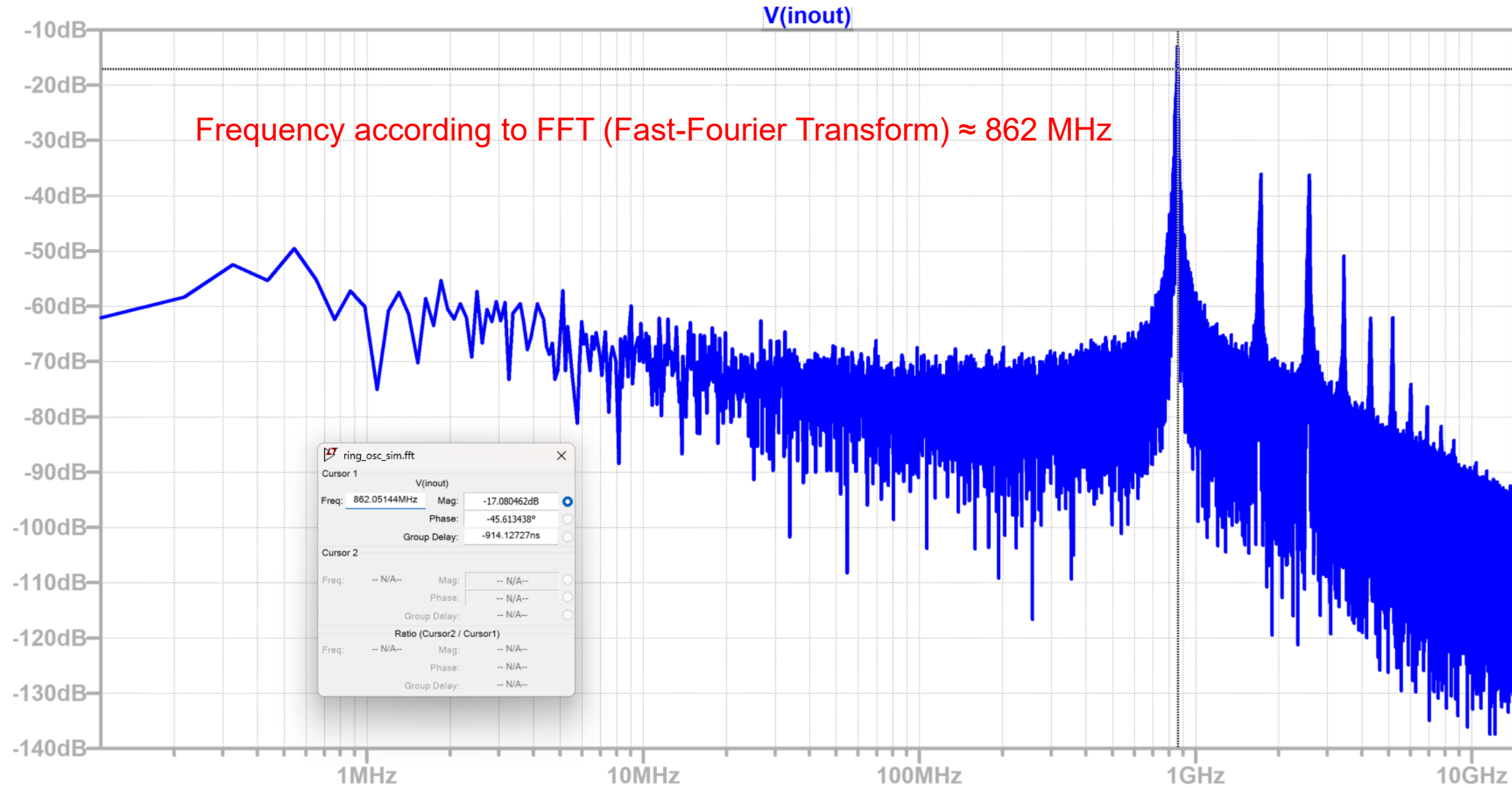
Each CMOS inverter has a delay of approximately $1.16 \text{ ns} / 6 = 0.193 \text{ ns}$

CMOS RING OSCILLATORS

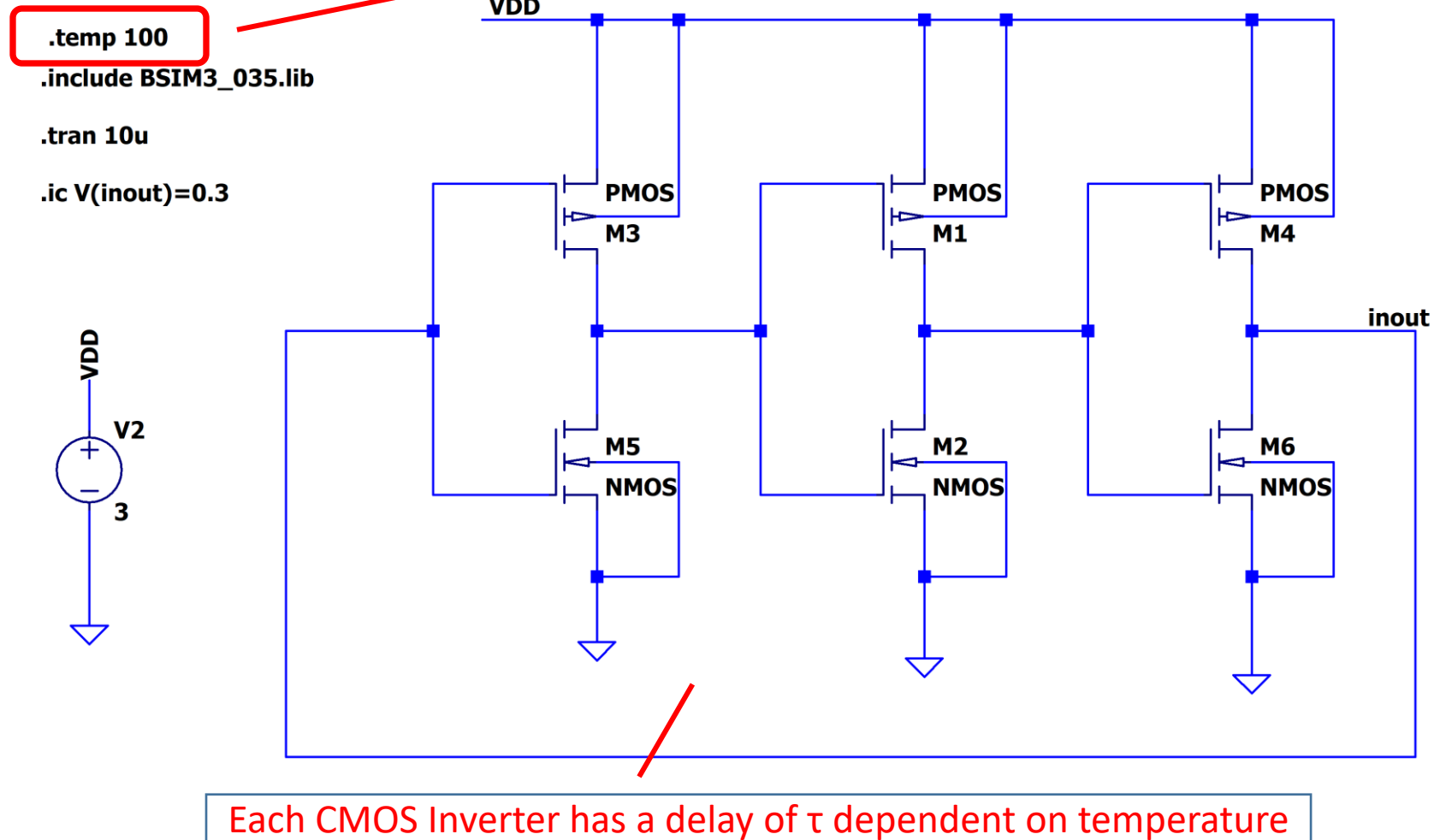


Perform a FFT and find the oscillation frequency in frequency domain

CMOS RING OSCILLATORS

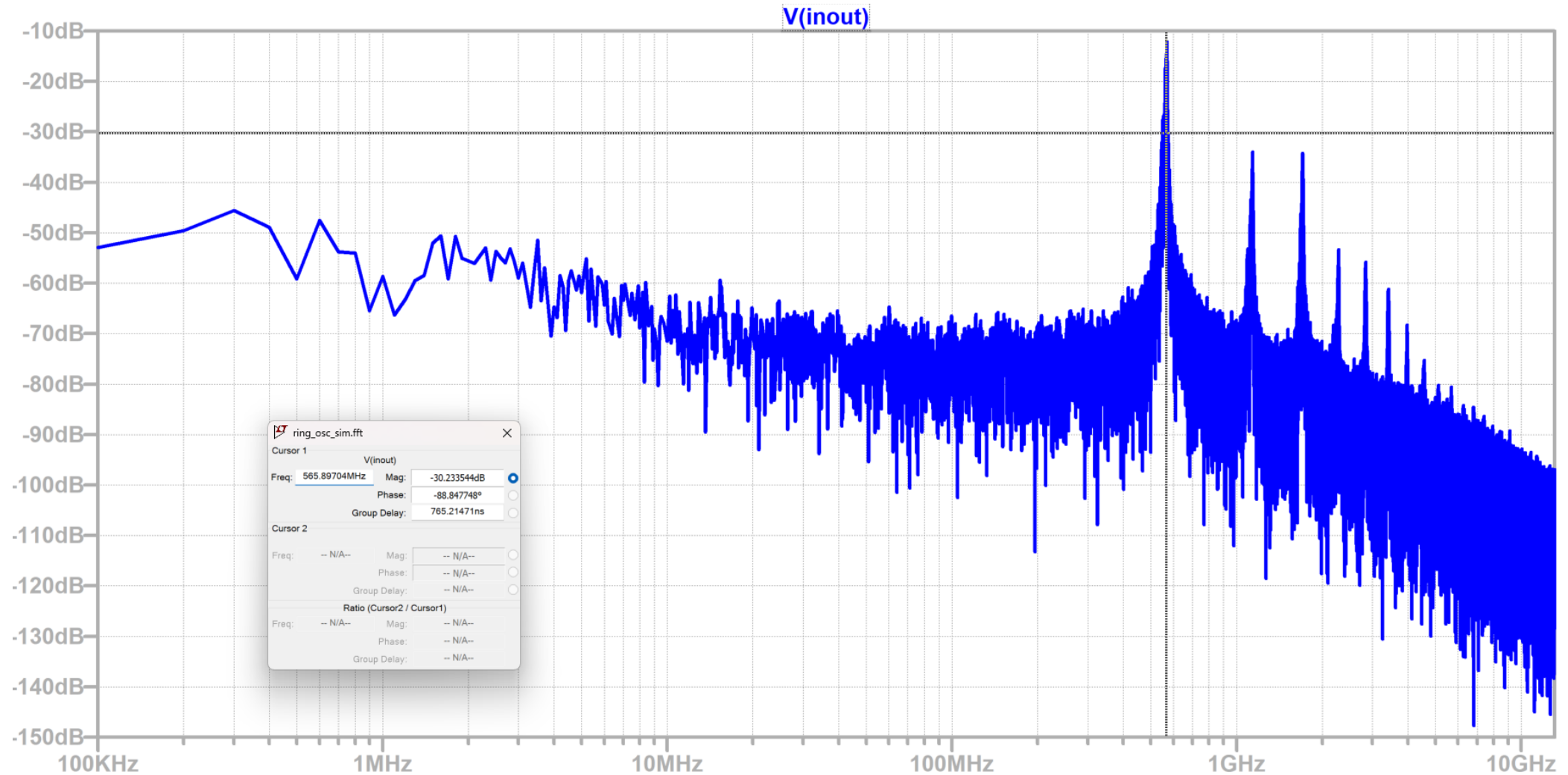


CMOS RING OSCILLATORS



If the temperature changes, the internal resistances ($R_{ds,on}$) of nmos and pmos transistors change which lead to a change in τ .

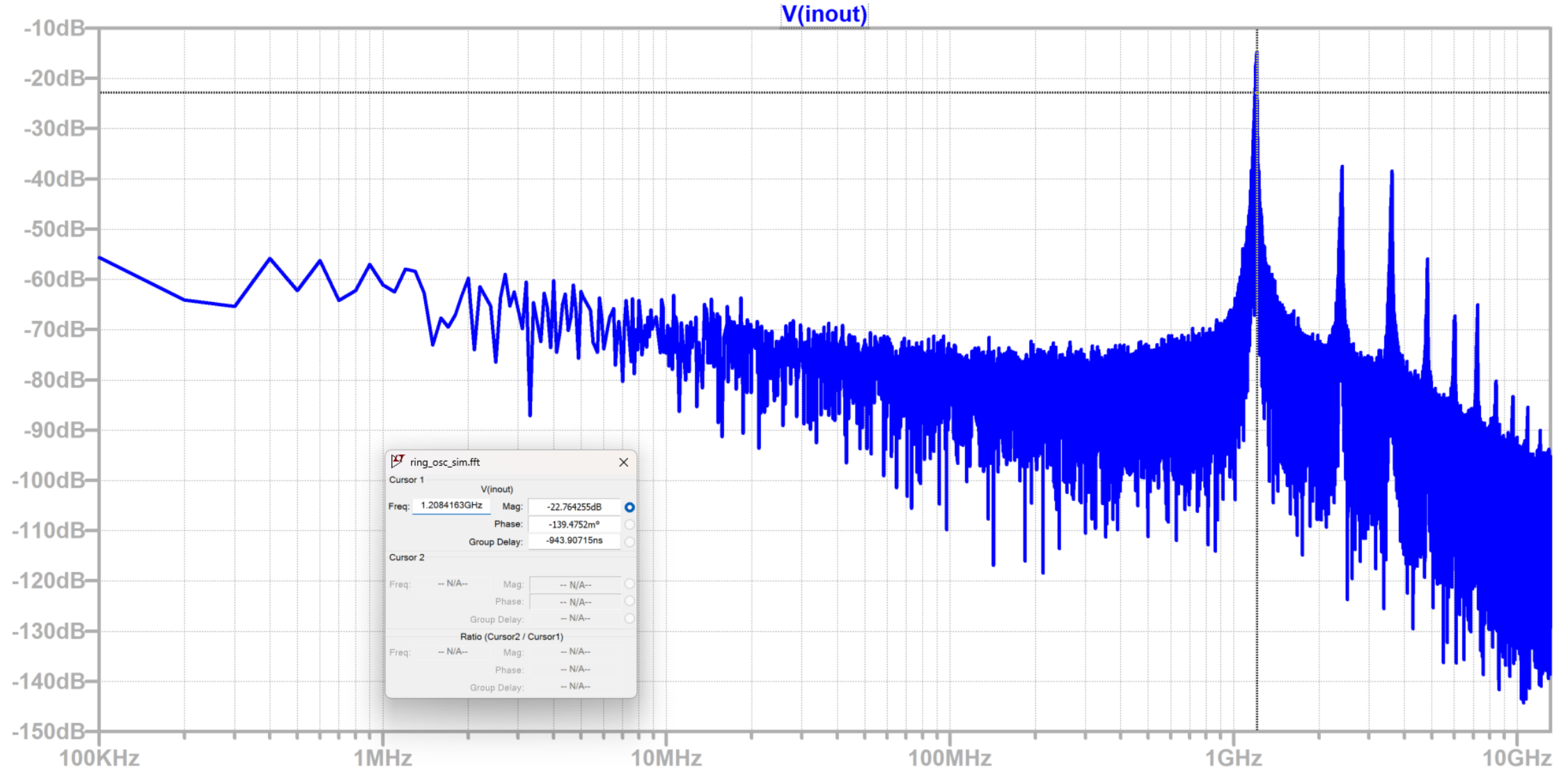
CMOS RING OSCILLATORS



Frequency = 560 MHz! → Frequency much lower at higher temperatures due to a lower mobility of electrons and therefore higher $R_{ds,on}$!

CMOS RING OSCILLATORS

Temperature = -40°C



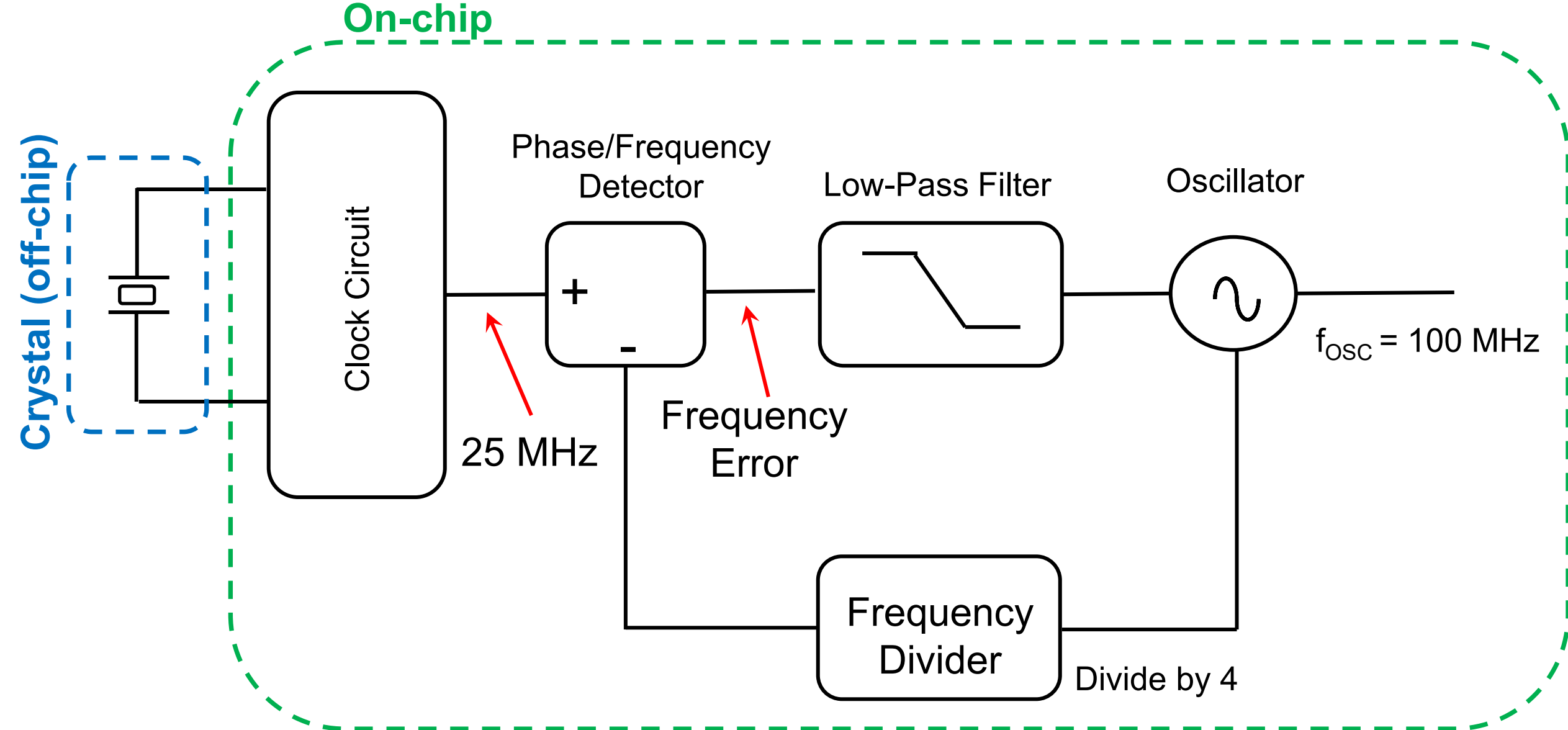
Frequency = 1200 MHz! → Frequency much higher at lower temperatures due to a higher mobility of electrons and therefore lower $R_{ds,on}$!

OSCILLATORS AND PHASE-LOCKED LOOPS (PLL)

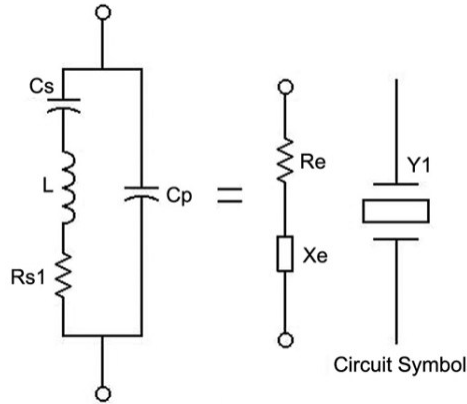
- ❖ There are also other types of oscillators besides Ring Oscillators: e.g., Colpitts Oscillators, Hartley Oscillators, Cross-Coupled Oscillators etc. The working principles of these oscillators will not be covered within the scope of this lecture.
- ❖ How do we compensate the temperature drift of oscillators? → Phase-Locked Loops (PLLs)

PLL

Example: Stable 100 MHz output by means of 25 MHz Clock Reference

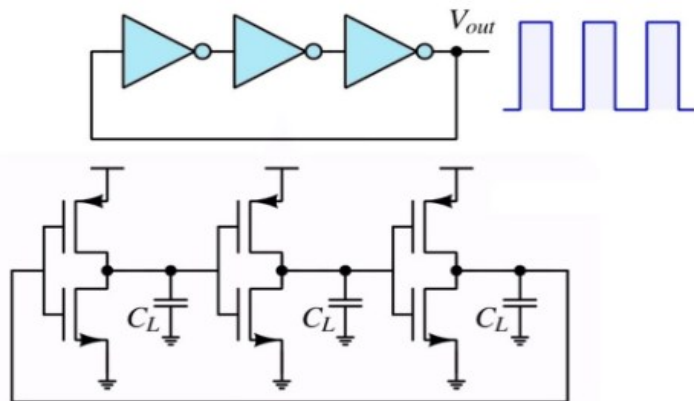


PLL

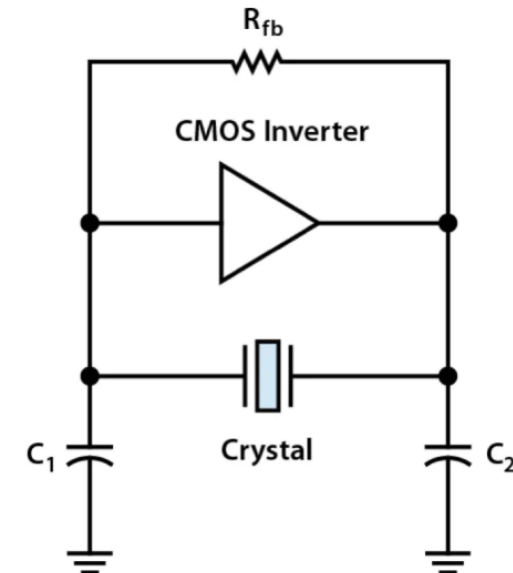


The crystal acts as a LC resonant tank and generates a stable sine wave vs. temperature together with the clock circuit

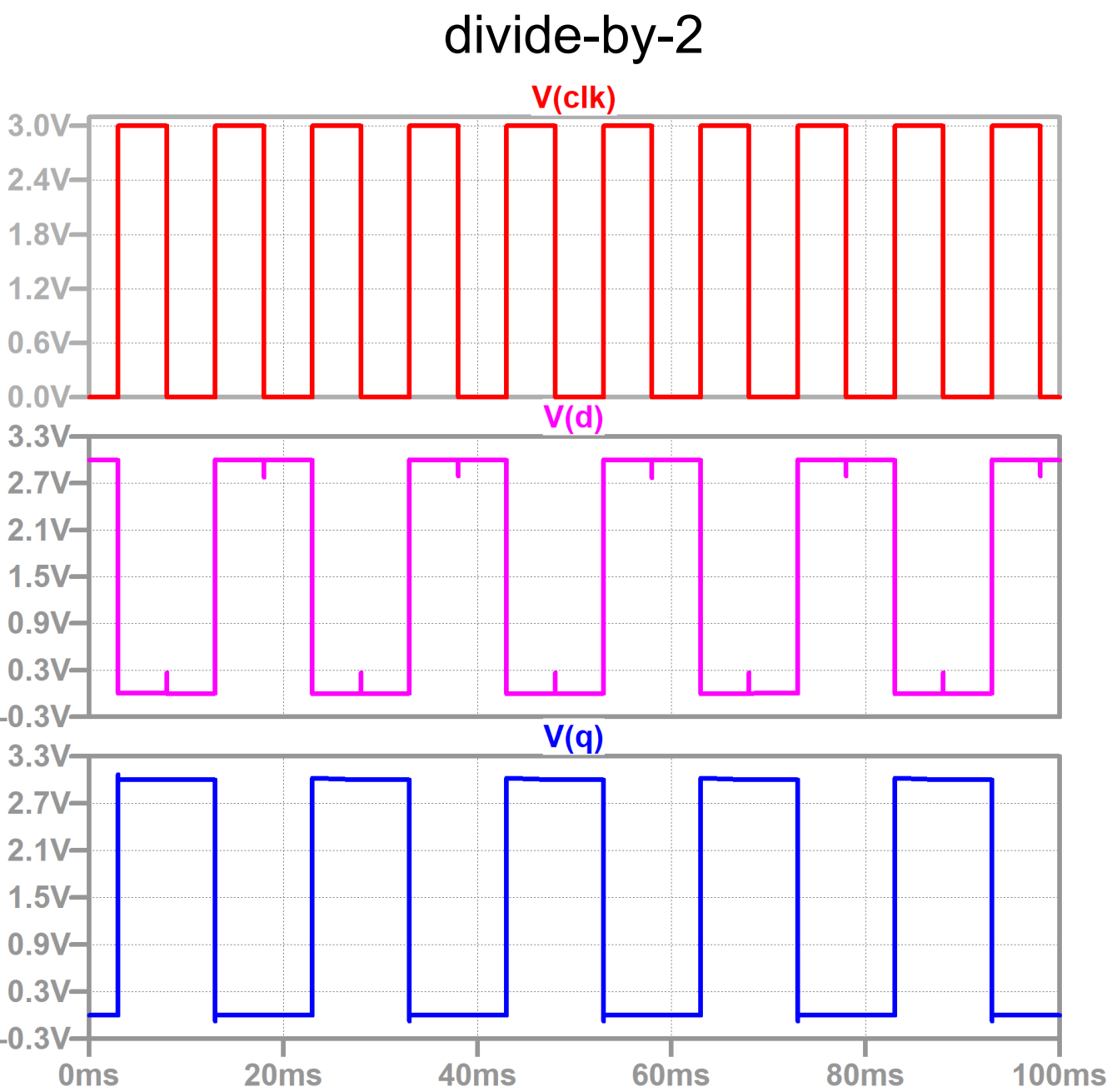
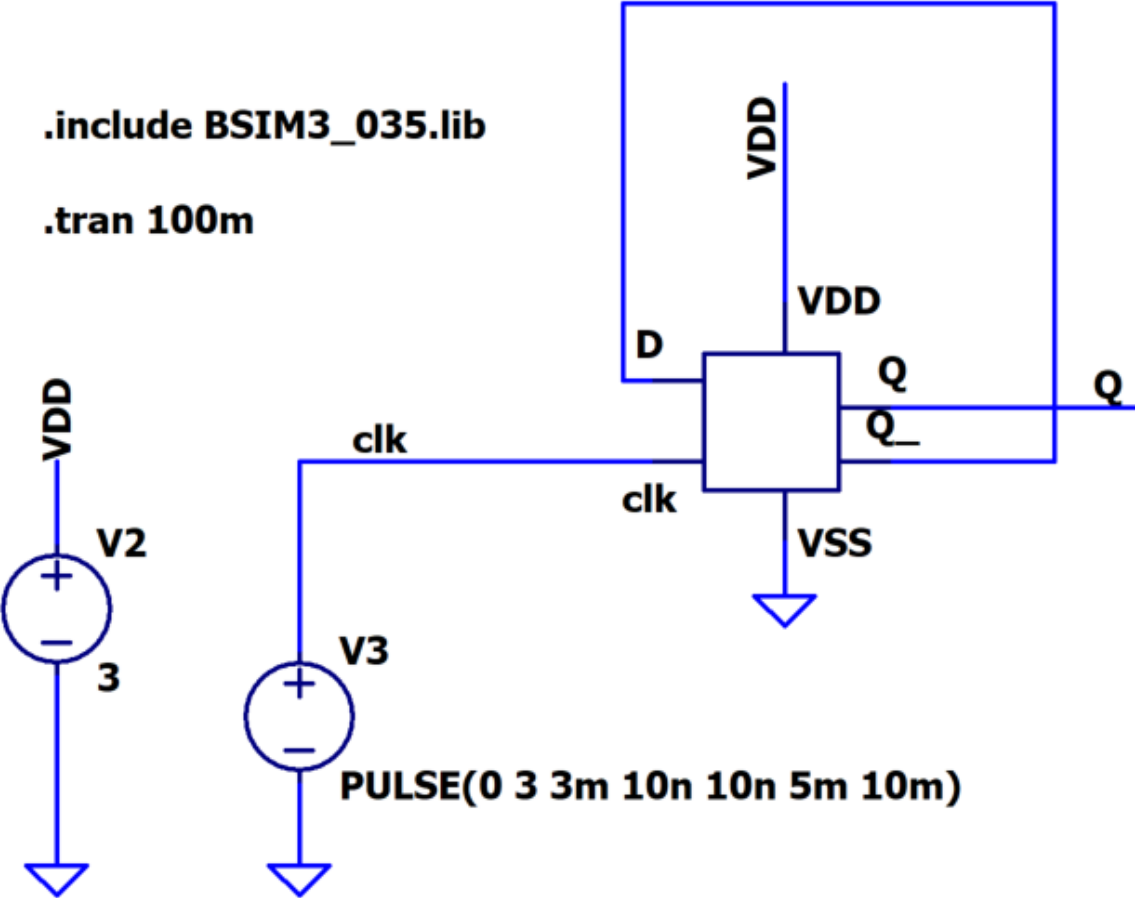
The difference between the oscillation frequency at the output of the frequency divider and the oscillation frequency at the output of the clock circuit is filtered and fed to the oscillator (e.g. ring oscillator). The ring oscillator frequency is adjusted (by for example using internal capacitors at the output of each CMOS inverter) such that the error becomes zero.



C_L is adjustable or in other words can be changed e.g., by using multiple capacitors switched in parallel



FREQUENCY DIVIDERS



References

1) www.halbleiter.org

The Principle of Semiconductor, Tokyo Electron Ltd.

2) Paul Gray, Robert Meyer, "Analysis and Design of Analog Integrated Circuits", John Wiley and Sons

3) Phillip E. Allen, Douglas R. Holberg, "CMOS Analog Circuit Design", Oxford University Press, 2002.

4) Texas Instruments Application Note, "Choose the right A/D converter for your application".